



Housing Recovery and Reconstruction Platform-Nepal (HRRP)

Orientation Program

on

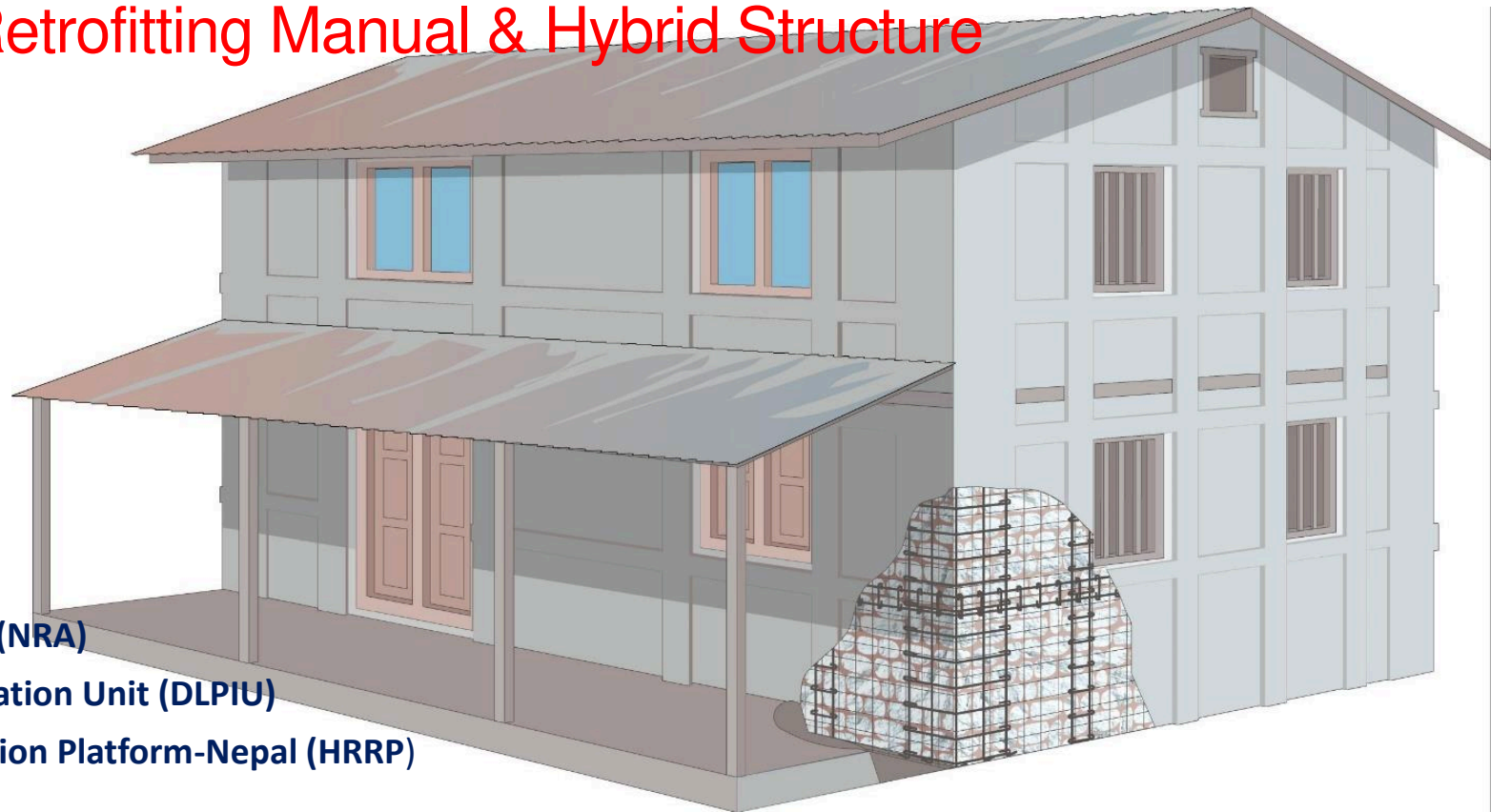
Repair & Retrofitting Manual & Hybrid Structure

Organized by:

National Reconstruction Authority(NRA)

MOUD/District Project Implementation Unit (DLPIU)

Housing Recovery and Reconstruction Platform-Nepal (HRRP)



Agenda



Housing Recovery and Recovery and Reconstruction Platform-Nepal
An orientation Program on “Repair & Retrofitting Manual & Hybrid Structure Manual”

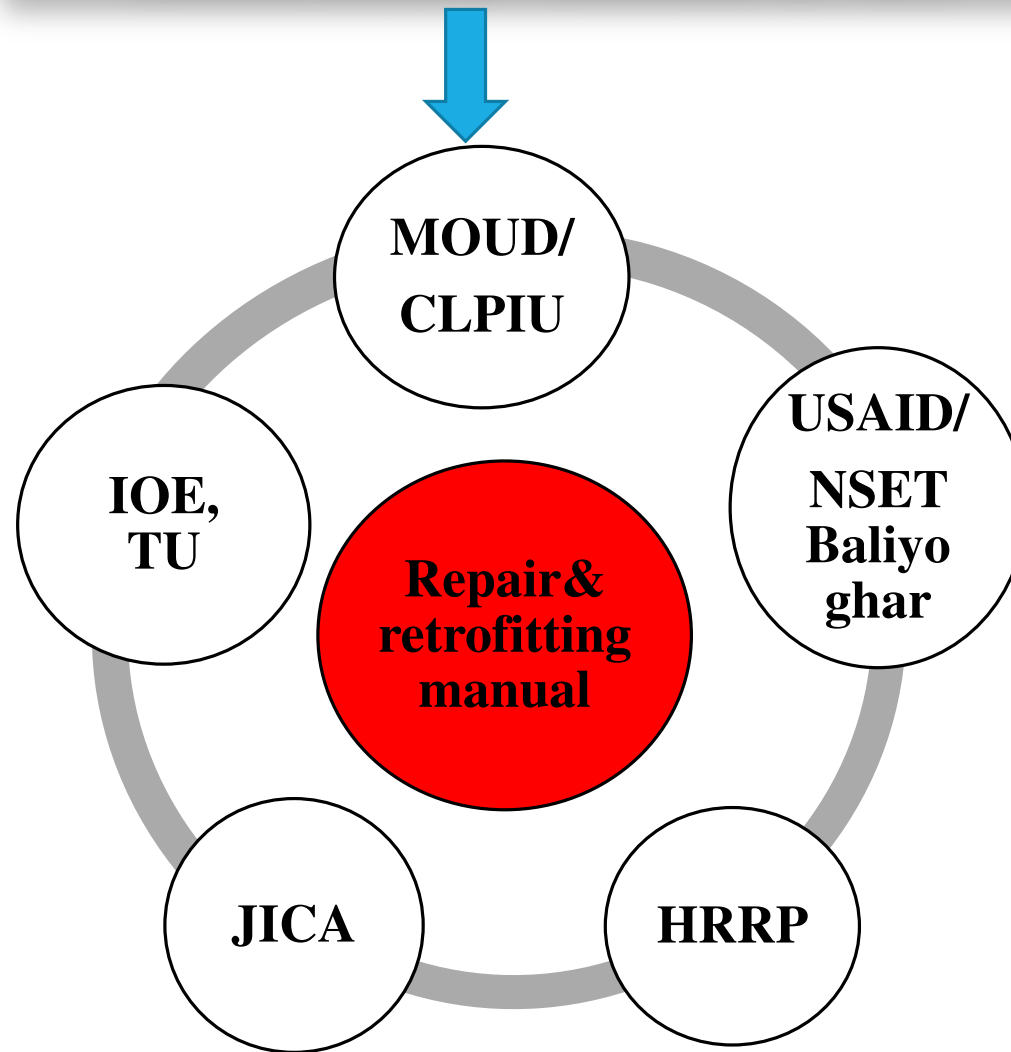
Schedule

Time		Session	Description
10:00	10:15	S0	Arrival/Registration/Opening Session
10:15	11:45	S1	Repair & Retrofitting Manual(Masonry & RCC Structure) • Introduction • Objectives & Scope of Manual • Inspection Method • Technical details of Repair & Retrofitting method ➤ Seismic Damage and Intervention ➤ Seismic Deficiencies and Intervention ➤ Ready to use seismic retrofit designs ➤ Construction sequences ➤ Typical structural drawing
11:45	12:15		Lunch
12:15	1:15	S2	Hybrid Structure Manual • Background and Characteristics of Hybrid Structure ➤ Background ➤ Definition ➤ Limitation ➤ Failure pattern ➤ Timing of inspection • Technical Specification of Hybrid Structure
1:15	2:00	S3	Discussion

- Almost 25,000 households across the 31 earthquake affected district were categorized as damage grade 2 of major technical solution or damage grade 3 of minor technical solution
- Under the Government of Nepal (GoN) housing reconstruction program, there is provision of housing retrofit grant of 100,000 NPRs if it complies with the relevant standards and specifications
- This manual outlines these standards and specifications as well as the minimum intervention works required to carry out the retrofitting

- To set the minimum criteria to provide the tranches under the retrofitting grant
- Cover policies for distribution of tranches with minimum technical intervention options
- To support engineers for inspection, help them to provide advice and guidance to households

Standardization Committee of NRA



Scope of Manual

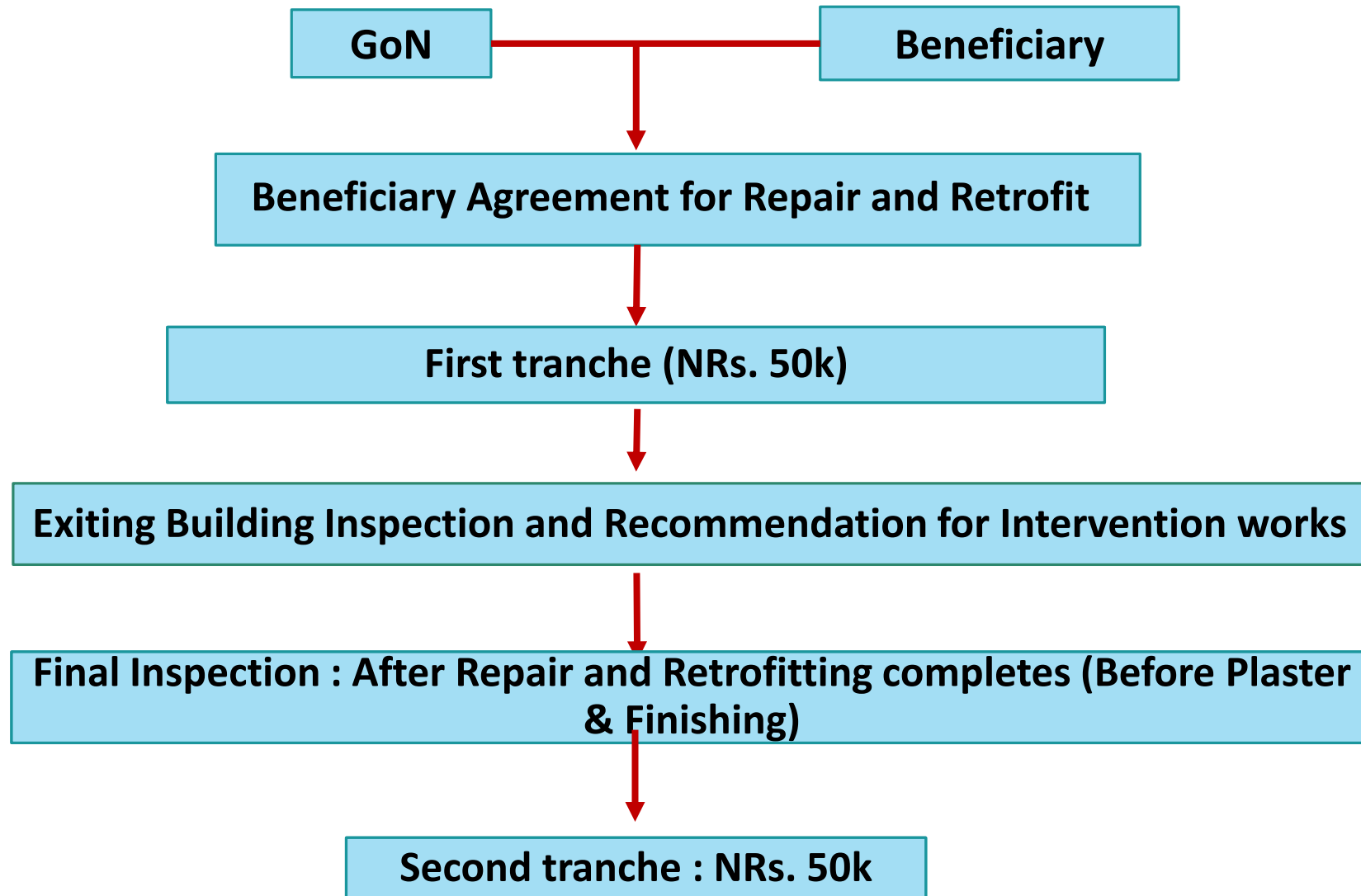
- The repair and retrofitting strategies set forth in this manual are applicable only for residential houses categorized as damage grade 2 (major) or 3 (minor) after Gorkha earthquake 2015.
- The designs mentioned in the manual are ready-to-use designs for all structural components, but some provisions mentioned are set as advisory measures.

- **Category “A”**: Modern building to be built, based on the international state-of-the-art, also in pursuance of the building codes to be followed in developed countries.
- **Category “B”**: Buildings with plinth area of more than one Thousand square feet, with more than Three floors including the ground floor or with structural span of more than 4.5 meters.
- **Category “C”**: Buildings with plinth area of up to One Thousand square feet, with up to Three floors including the ground floor or with structural span of up to 4.5 meters.
- **Category “D”**: Small Houses, sheds made of baked or unbaked brick, Stone, Clay, bamboos, grass etc., except those set forth clause (a), (b) & (c)

- The repair and retrofitting strategies are only for damaged non-engineered residential buildings falls under fall under category 'C' and 'D' of NBC.

If the intervention has already been completed as per, or similar to, the strategies outlined in this manual, Government of Nepal published documents, or as per international practices, and are based on codal provision ensuring life safety with quality construction, then applications can be forwarded only after thorough engineering judgement

Grant Process



General Terminology

Repair:

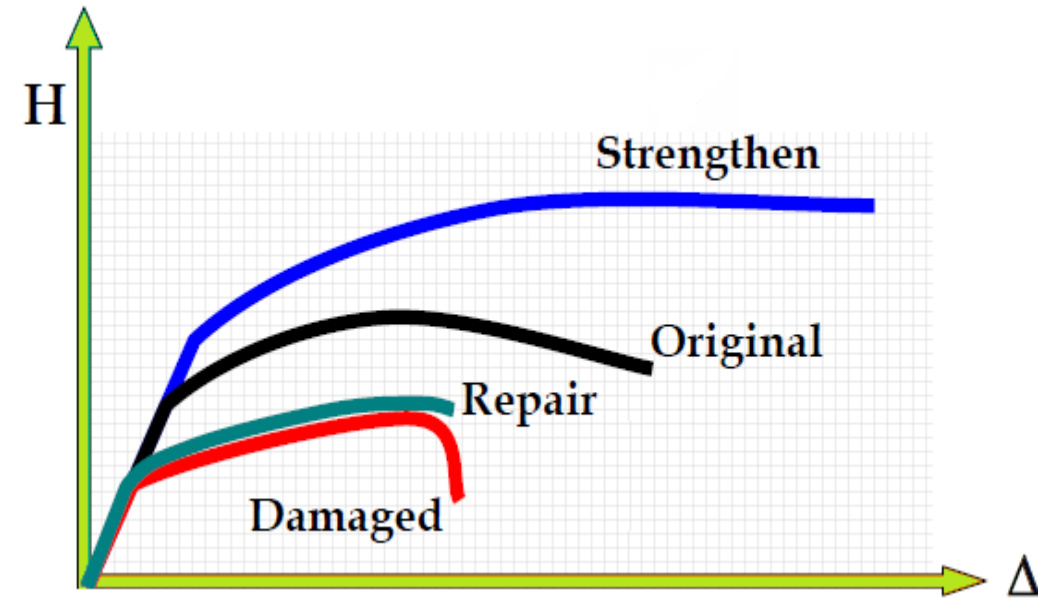
To bring back the architectural shape of the building so that all services start working & the functioning of building is resumed quickly.

Restoration:

To carry out structural repairs to load bearing components to restore its original strength.

Retrofitting:

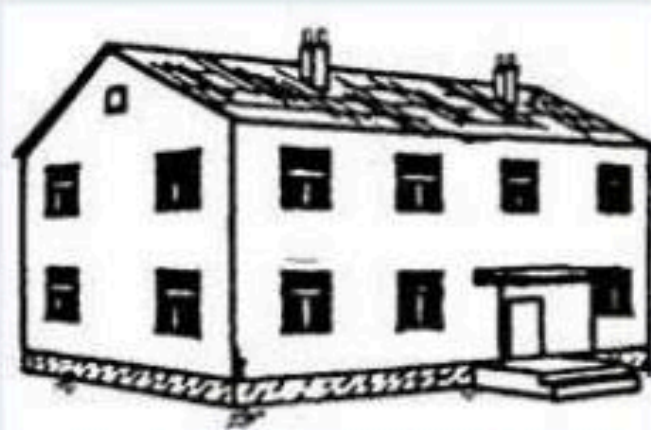
To make buildings stronger than before.



MASONRY STRUCTURE

Damage Grade: Masonry Buildings

Classification of damage to masonry buildings



Grade 1: Negligible to slight damage

Structural damage : No
Non-structural damage: Slight

- Hair-line cracks in very few walls.
- Fall of small pieces of plaster only.
- Fall of loose stones from upper parts of buildings in very few cases.



Grade 2: Moderate damage

Structural damage : Slight
Non-structural damage: Moderate

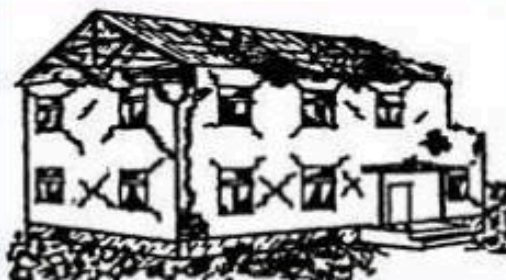
- Cracks in many walls.
- Fall of fairly large pieces of plaster.
- Partial collapse of chimneys.



Grade 3: Substantial to heavy damage

Structural damage: Moderate
Non-structural damage: Heavy

- Large and extensive cracks in most walls.
- Roof tiles detach.
- Chimneys fracture at the roof line; failure of individual non-structural elements (partitions, gable walls).



Grade 4: Very heavy damage

Structural damage: Heavy
Non-structural damage: Very heavy

- Serious failure of walls; partial structural failure of roofs and floors.

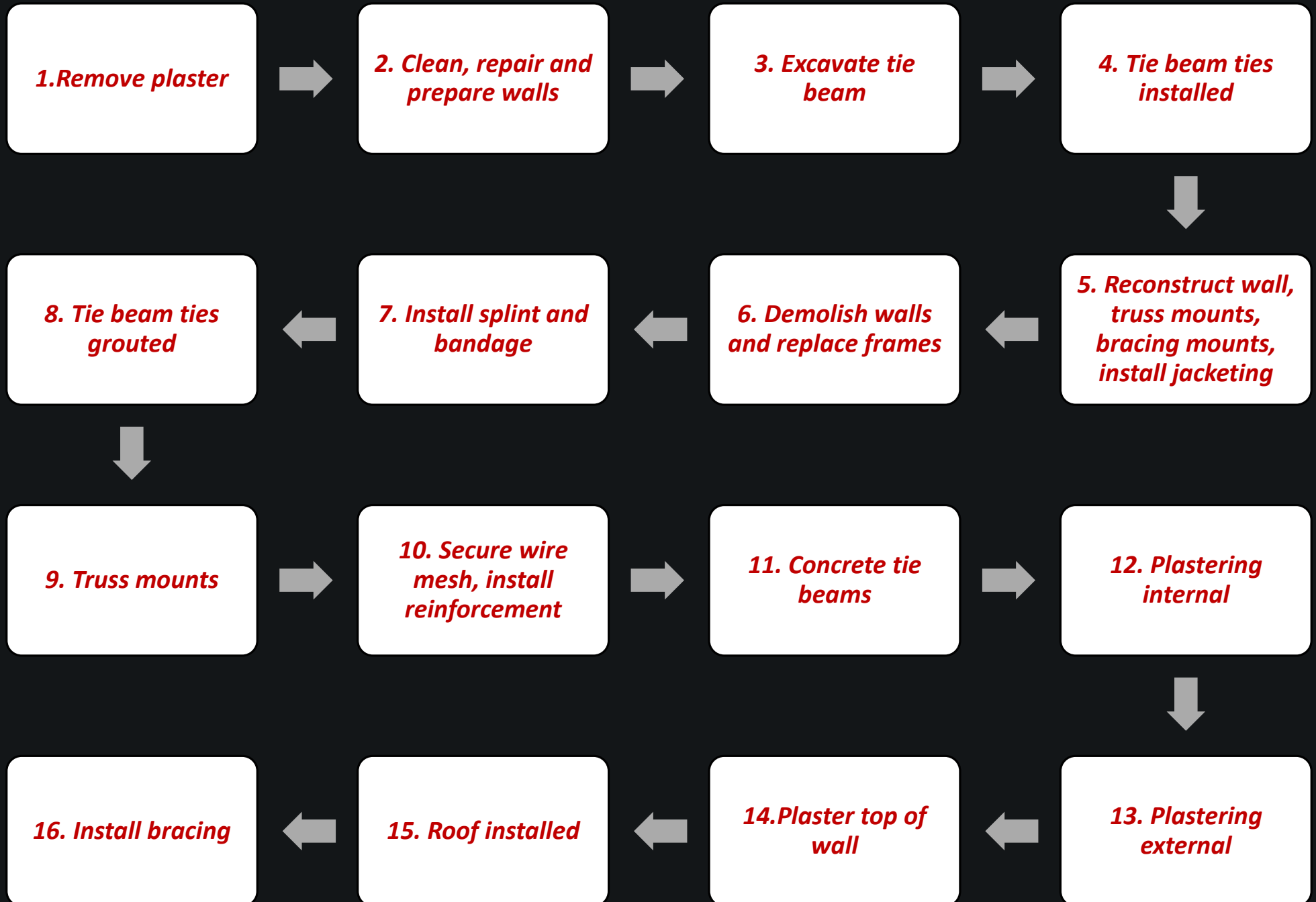


Grade 5: Destruction

Structural damage: very heavy

- Total or near total collapse.

Retrofitting process



PART-A: Seismic Damage and Intervention

PART-B: Seismic Deficiencies and Intervention

PART-C: Ready to use seismic retrofit designs

PART-D: Construction sequences

Annex 1 : Typical structural drawing

Annex 2 : EMS Damage Grade

PART-A: Seismic Damage and Intervention

PART-A: Seismic Damage and Intervention

This part deals with seismic damages and possible intervention that needs to turn the building in to pre-earthquake condition.

Its details in Part D Construction sequences

1. Foundation damage & mitigation work
[Key problem, repair solution and retrofit solution]
2. Roof/Floor partial collapse and Mitigation Work
[Key problem, repair solution and retrofit solution]
3. Cracks in wall and Mitigation Work
[Key problem, repair solution and retrofit solution]

1.Foundation damage & mitigation work

Potential damages in foundation are as follows :

- F.1 Cracked stone masonry
- F.2 Settlement of Foundation
- F.3 Bulged Stone Walls
- F.4 Dislocations and Loose Stone
- F.5 Stone foundation wall interruption, removal of portions of the wall, & loss of structural integrity

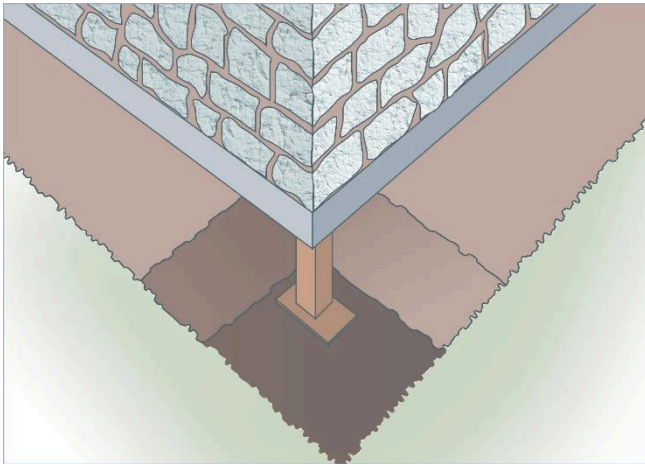
Foundation damage & mitigation work Contd.



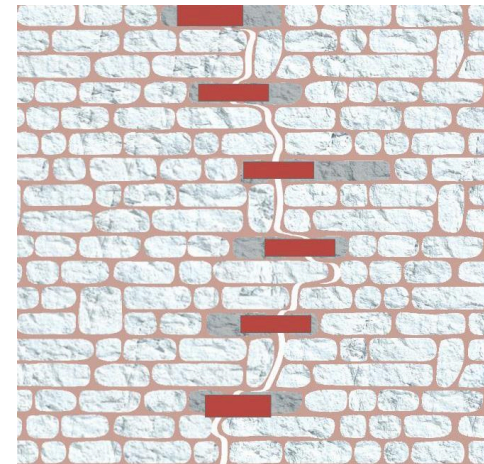
F.3 Bulged and leaning stone Foundation



F.2 Settlement of Foundation



Reconstruction of Foundation with proper safety



Repair Foundation

Roof/Floor partial collapse and Mitigation Work



Potential damages in floor and/or roof are as follows :

- R.1: Partial to heavy damage on gable wall
- R.2 : Sliding of roof materials (stone slate or clay tiles)
- R.3 : Roof connections failure
- R.4 : Floor connection
- R.5 : Floor to wall connection
- R.6 : Roof to wall connection
- R.7 :Wall to wall connection

R1. Partial to heavy damage of Gable wall



R1.1 Gable wall toppling



R.1.2 Gable wall toppling

Solution : Remove heavy material replace with the lighter materials



Use of C.G.I. sheet at Gable part

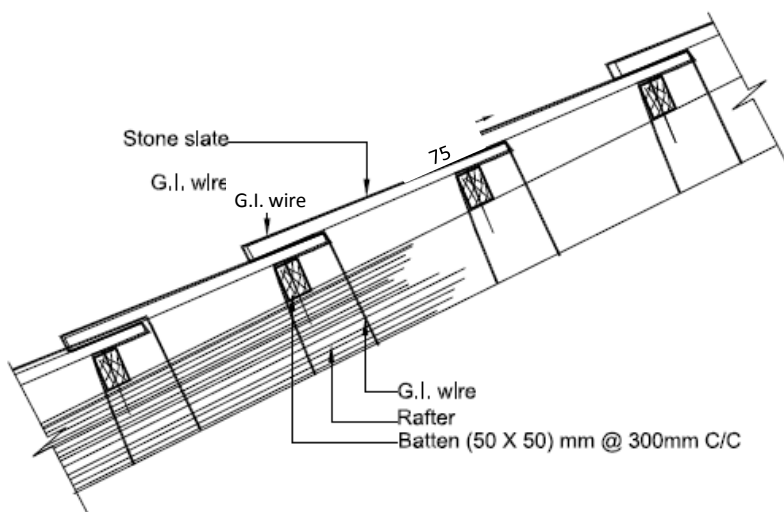


Use of timber planks at Gable part

R.2:Sliding of Roofing materials



Solution



Detail of anchoring slate stone

Source: NBC203 2015, P44



Detail of fixing clay tile

Source: NBC203 2015, P45

Detail of fixing clay tile

Source: NBC203 2015, P45

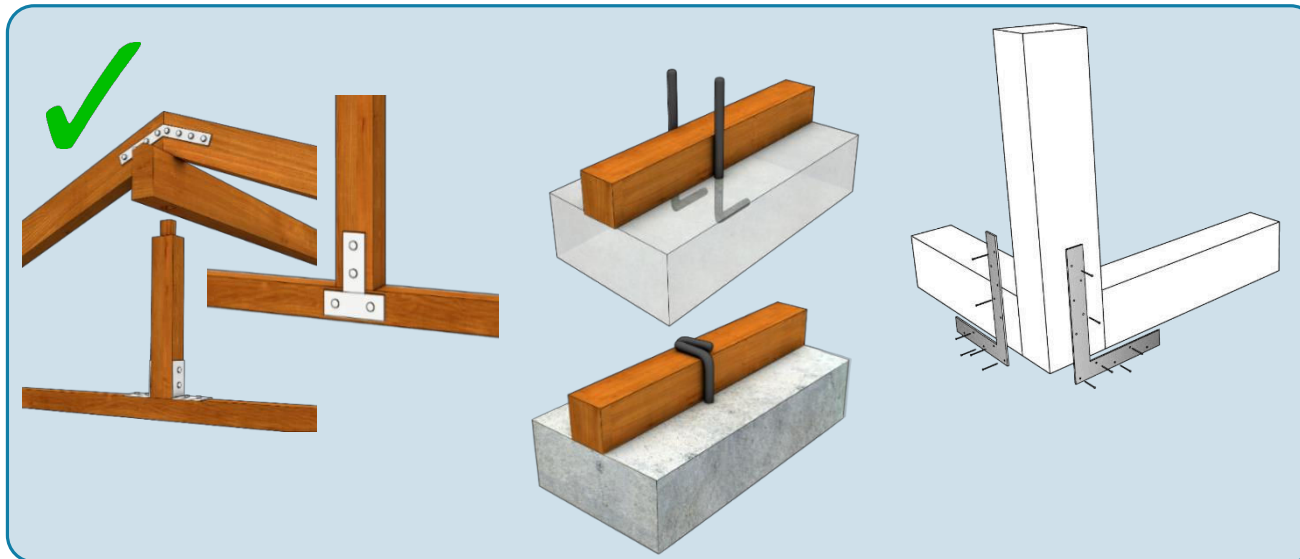
R.3:Roof connection failure



R.3.1 Purlin detached from rafter due to inadequate nailing



R.3.2 Connection of wooden truss



R.4: Floor connection



rafters R.4.1 view inside the attic showing the and the absence of collar beams and joist connections to the rafters

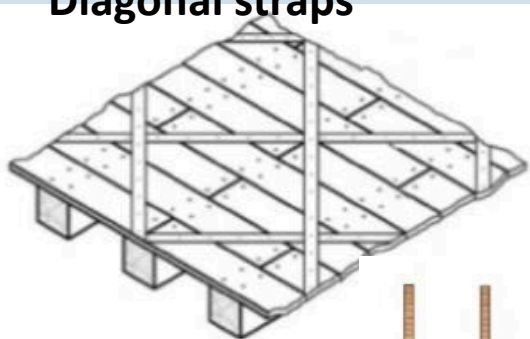


R.4.2 Inadequate and poor floor members

Solution



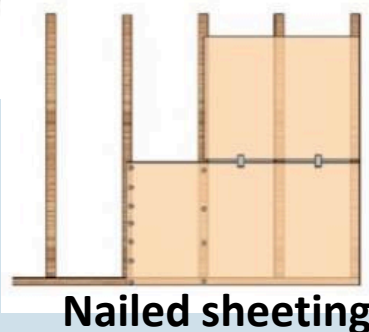
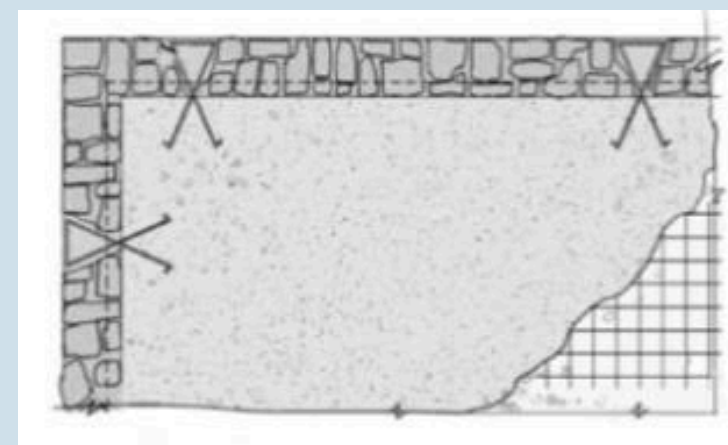
Diagonal straps



Connectivity of roof member



New RC Floor



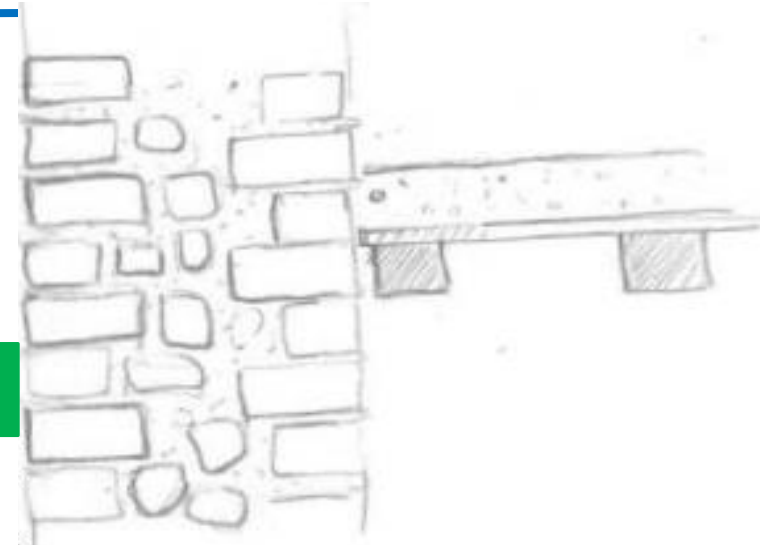
Nailed sheeting

R.5: Floor to wall connection



- R.5.1 Floor joists supported by half of the wall width (insufficient anchorage)

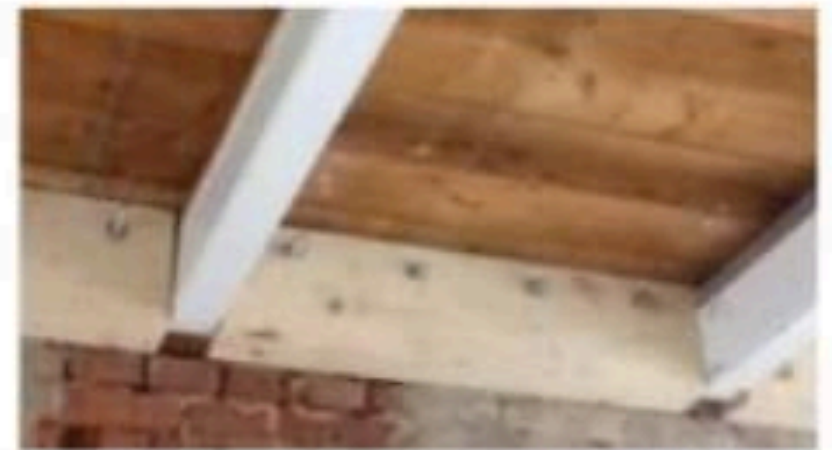
Solution



- R.5.2 Wall-to-floor connection parallel to the joists (insufficient anchorage)



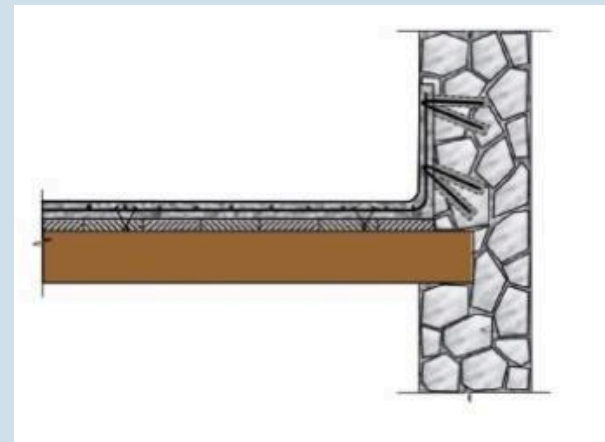
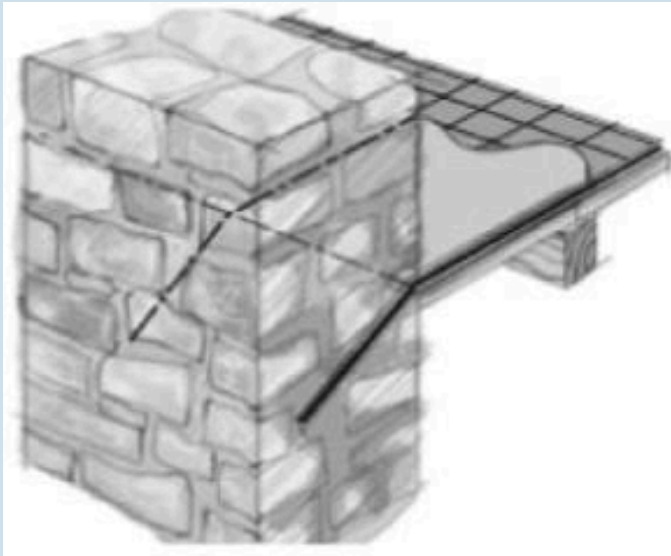
**Anchored with
splint and bandage
at floor level**



R.6:Roof to wall connection



Solution



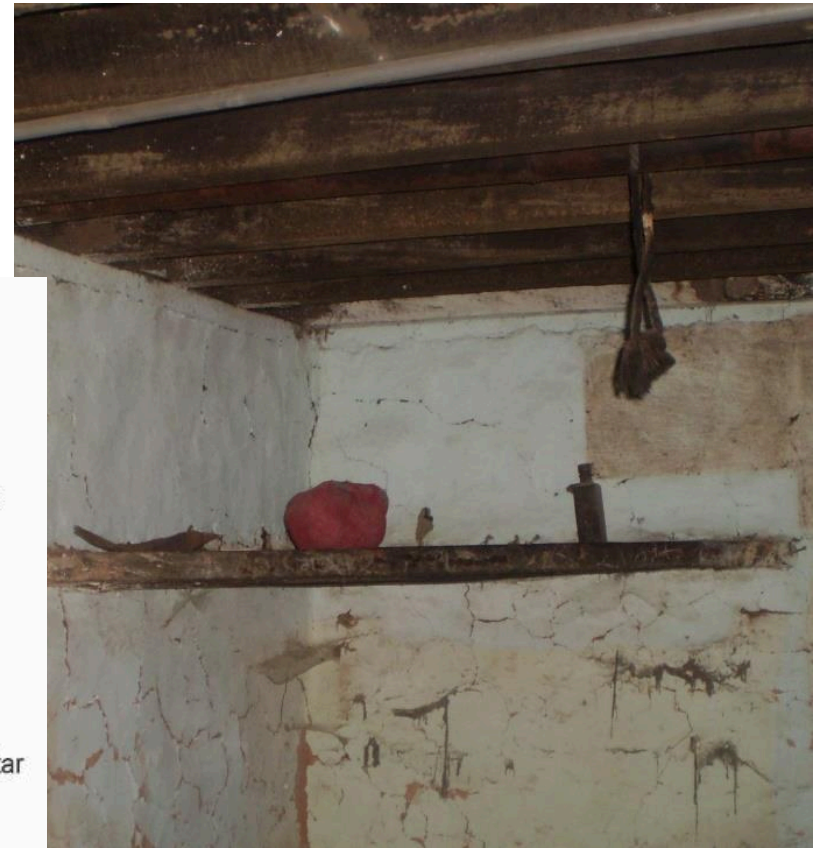
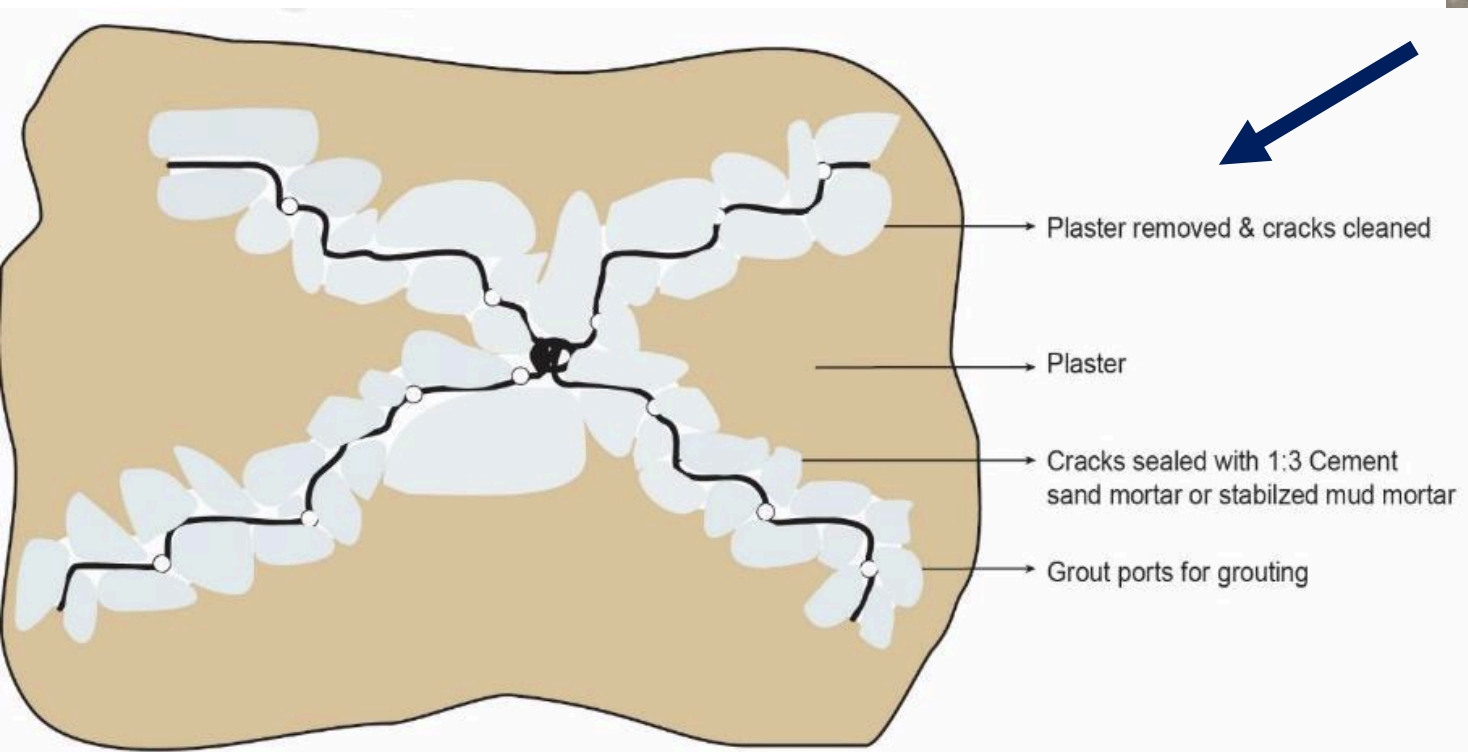
3. Cracks in wall and mitigation works



Potential damages in structural and non-structural wall are as follows :

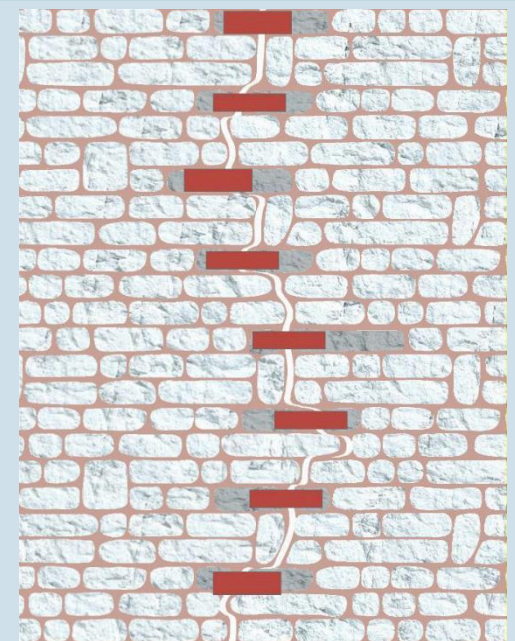
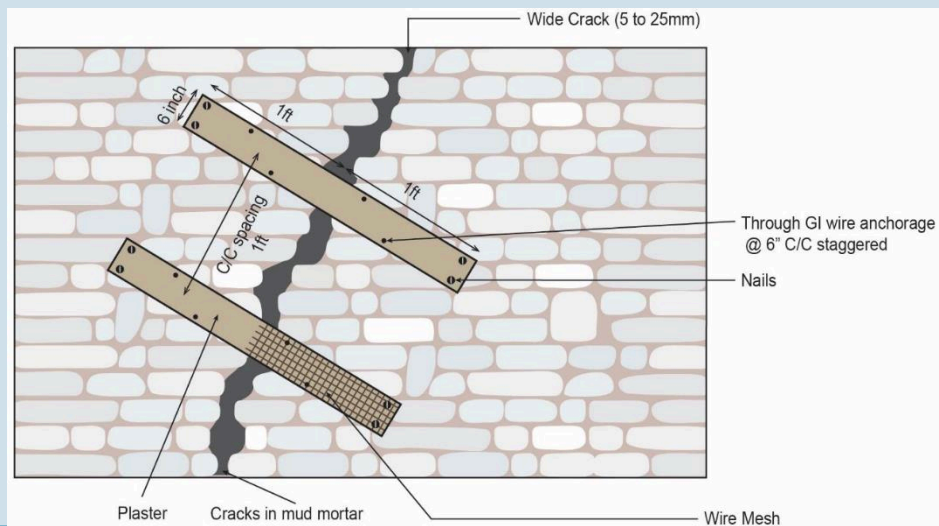
- C.1 : Minor cracks
- C.2 : Major cracks
- C.3 : Heavy cracks
- C.4 : Out-of-plane failure of walls
- C.5 : Wall to wall connection

C.1: Minor Cracks

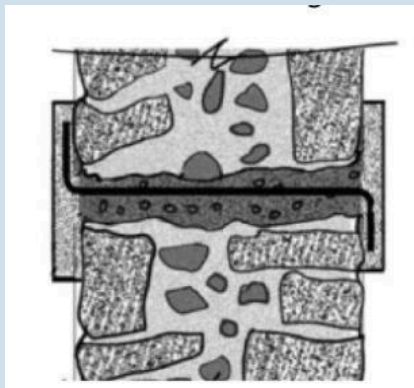


Repair Minor To Cracks Using Grouting

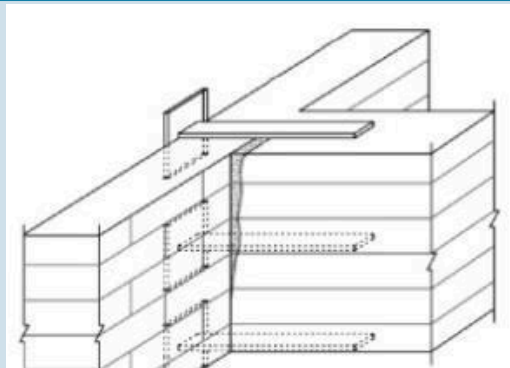
Major to heavy Crack



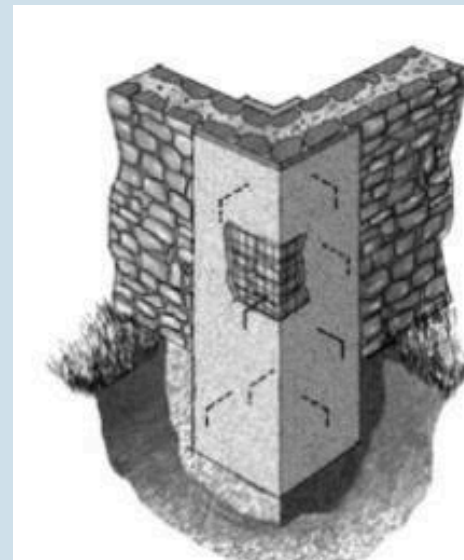
Wall to wall connection



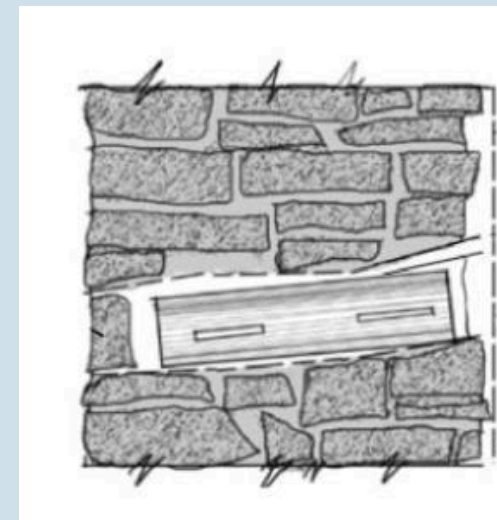
Anchored splint overlay



Metal stitching



Through wall anchor



Through wall anchor

PART-B: Seismic Deficiencies and Intervention

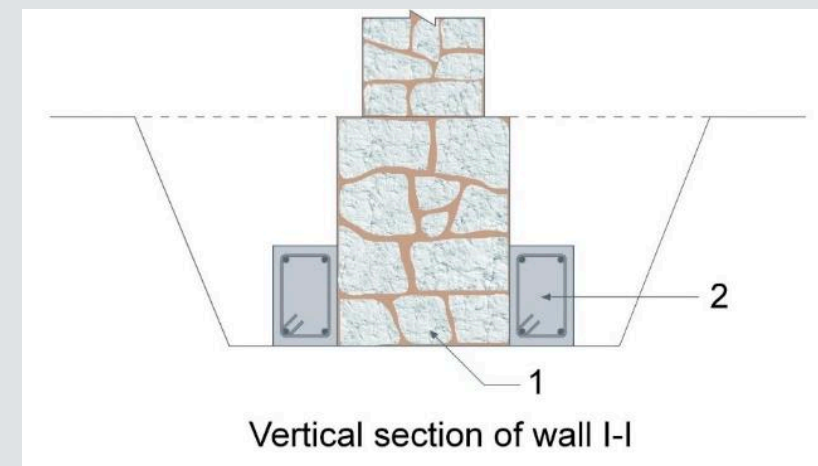
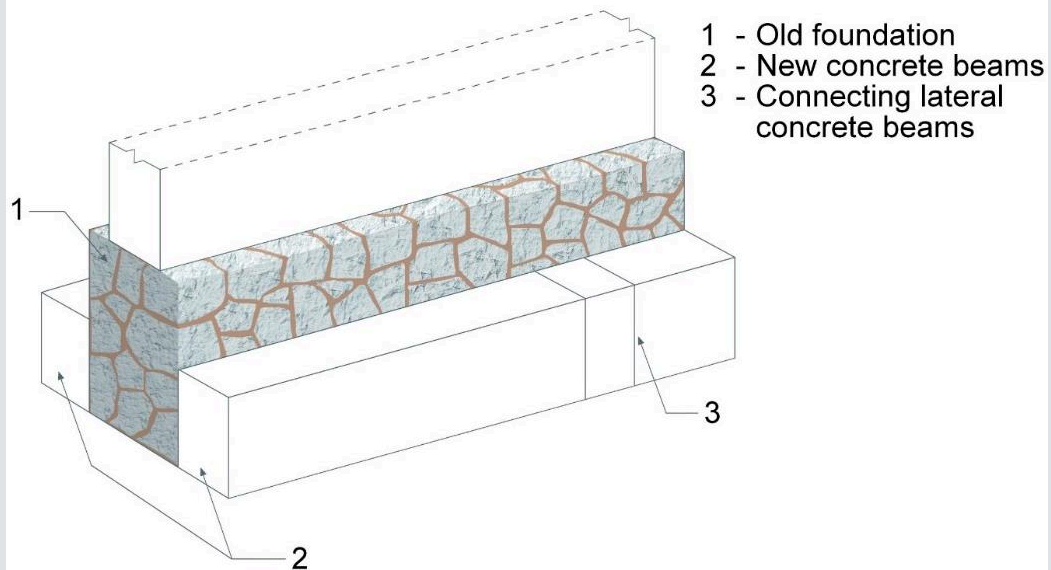
PART-B : Seismic deficiencies and intervention



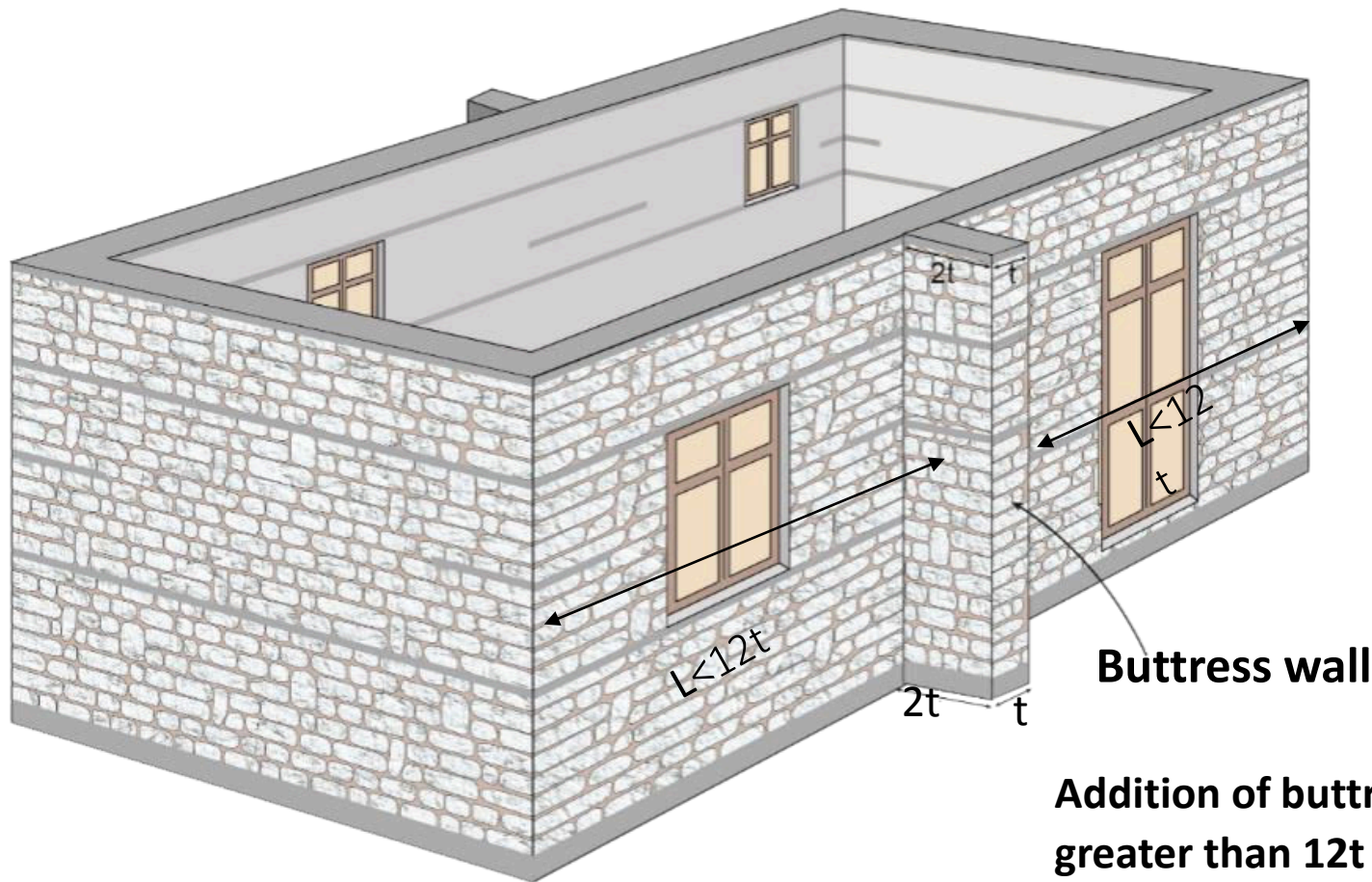
This part deals with possible deficiencies in the masonry buildings and possible improvement on. Probable intervention are as follows:

1. Foundation improvement 2. Configuration and load path improvement 3. Connection improvement between wall to wall 4. Connection improvement between wall to floor 5. Connection improvement between wall to roof	6. Stiffening of floor in their plane 7. Stiffening of roof in their plane 8. Tying of parapet wall 9. Tying of gable wall 10. Capacity improvement of structural wall with splint and bandage using:
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1. Foundation improvement

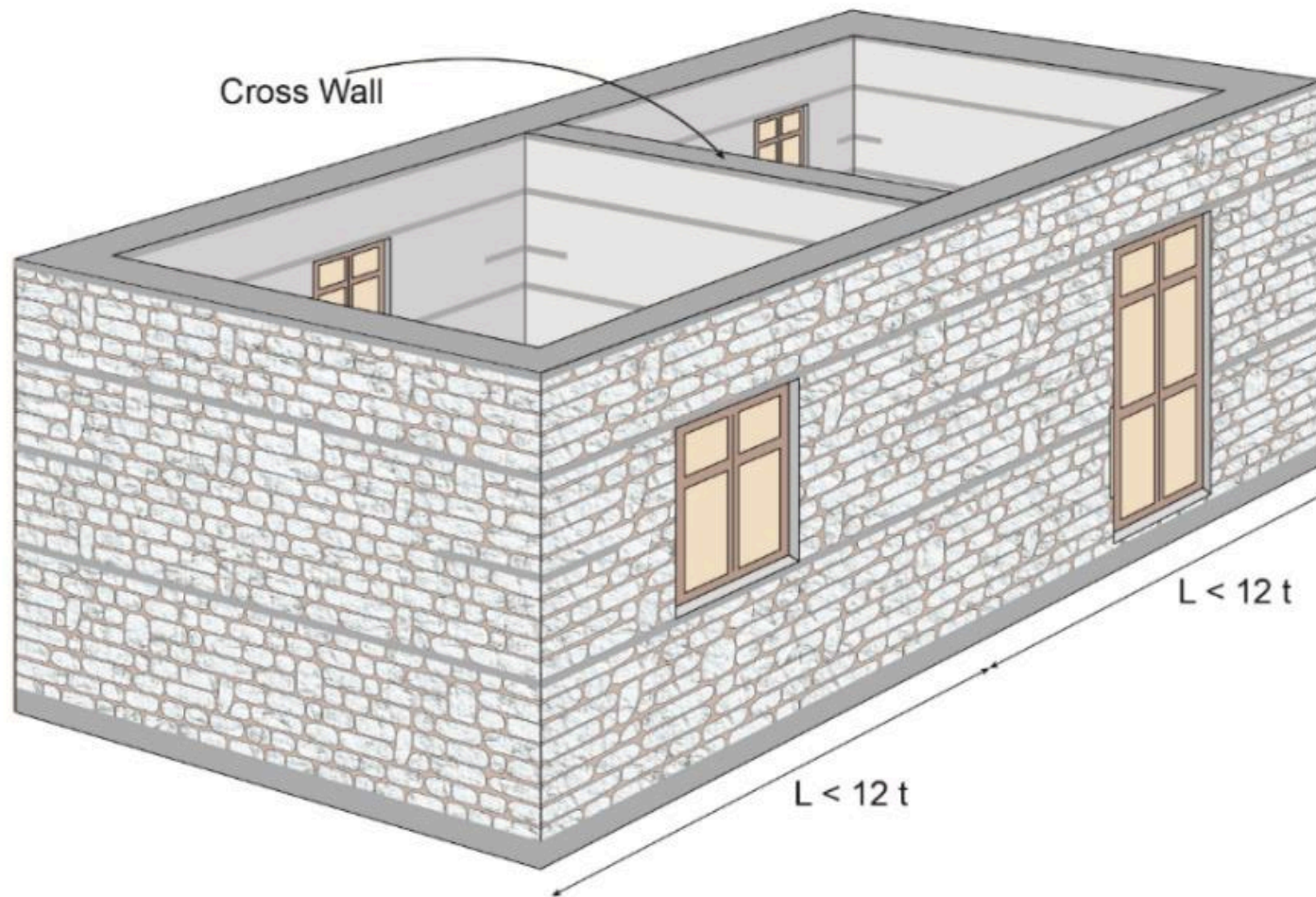


Example 1 : Construction details of Buttrressing in mud mortar with flexible floor



Addition of buttress wall when length of wall is greater than $12t$

Example 2 : Construction details of New Cross wall in mud mortar with flexible



Addition of cross wall when length of wall is greater than $12t$.

3.0 Connection improvement between wall - to – wall



Rebar anchored R C splint



**G I wire mesh anchored
R C splint**



3.0 Connection improvement between wall - to – wall

Installation of wooden vertical member from inside of the wall

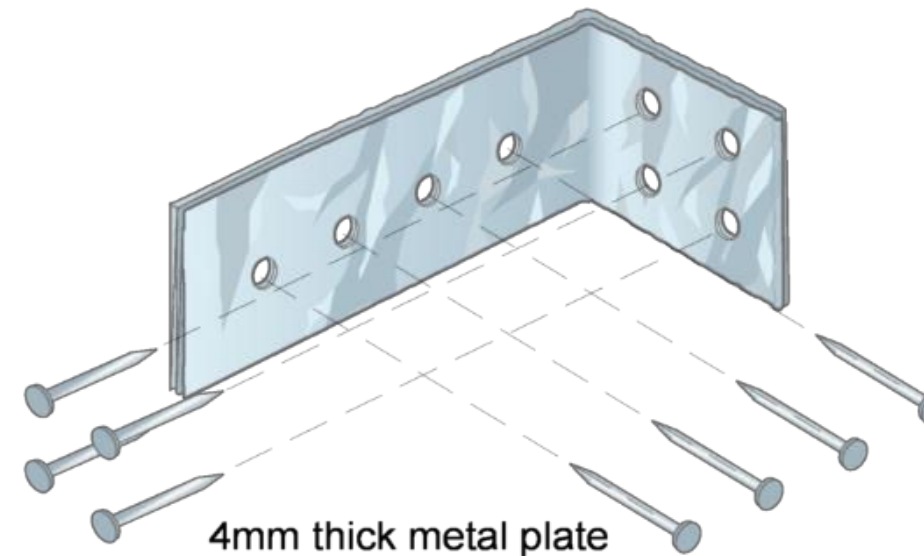


4.0 Connection improvement between wall-to- floor

When it is not possible to transport concrete and steel to the site:



Steel Strap

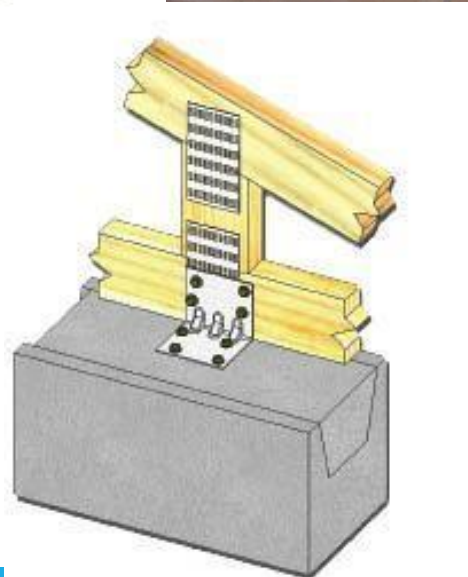


Details of metal strap

5.0 Connection improvement between wall - to - roof



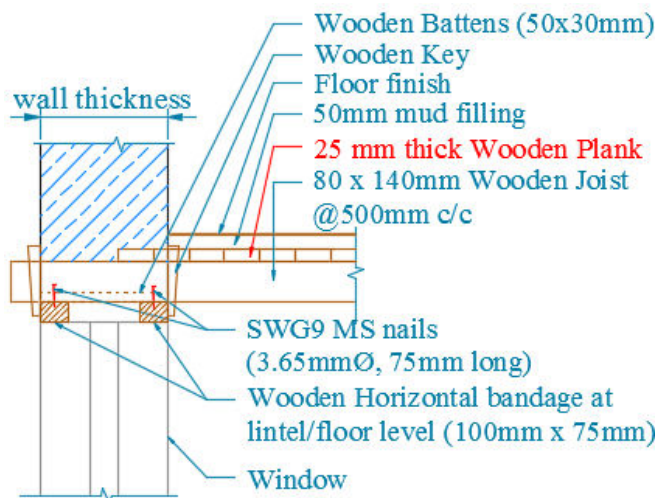
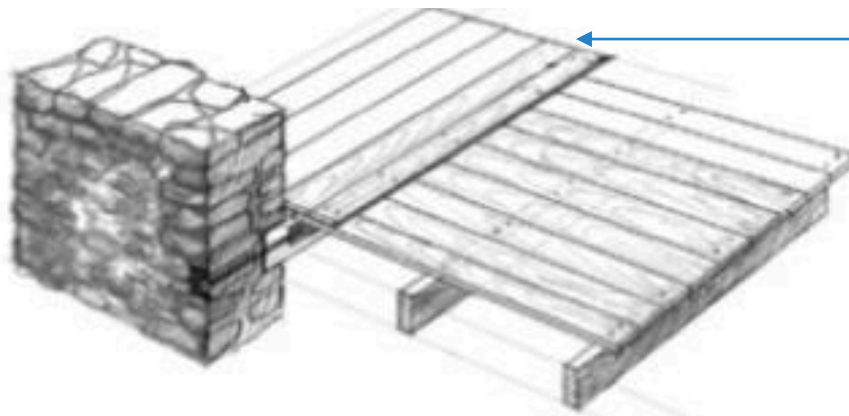
Option 1: Metal Strap at the top of wall:



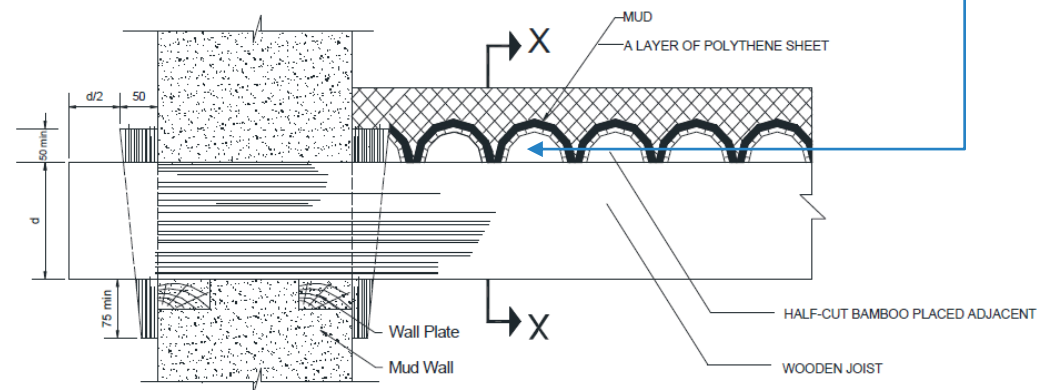
6.0 Stiffening of floors in their plane

Option 1: Wooden plank overlay or half cut bamboo overlay:

25mm thick Wooden Plank overlay

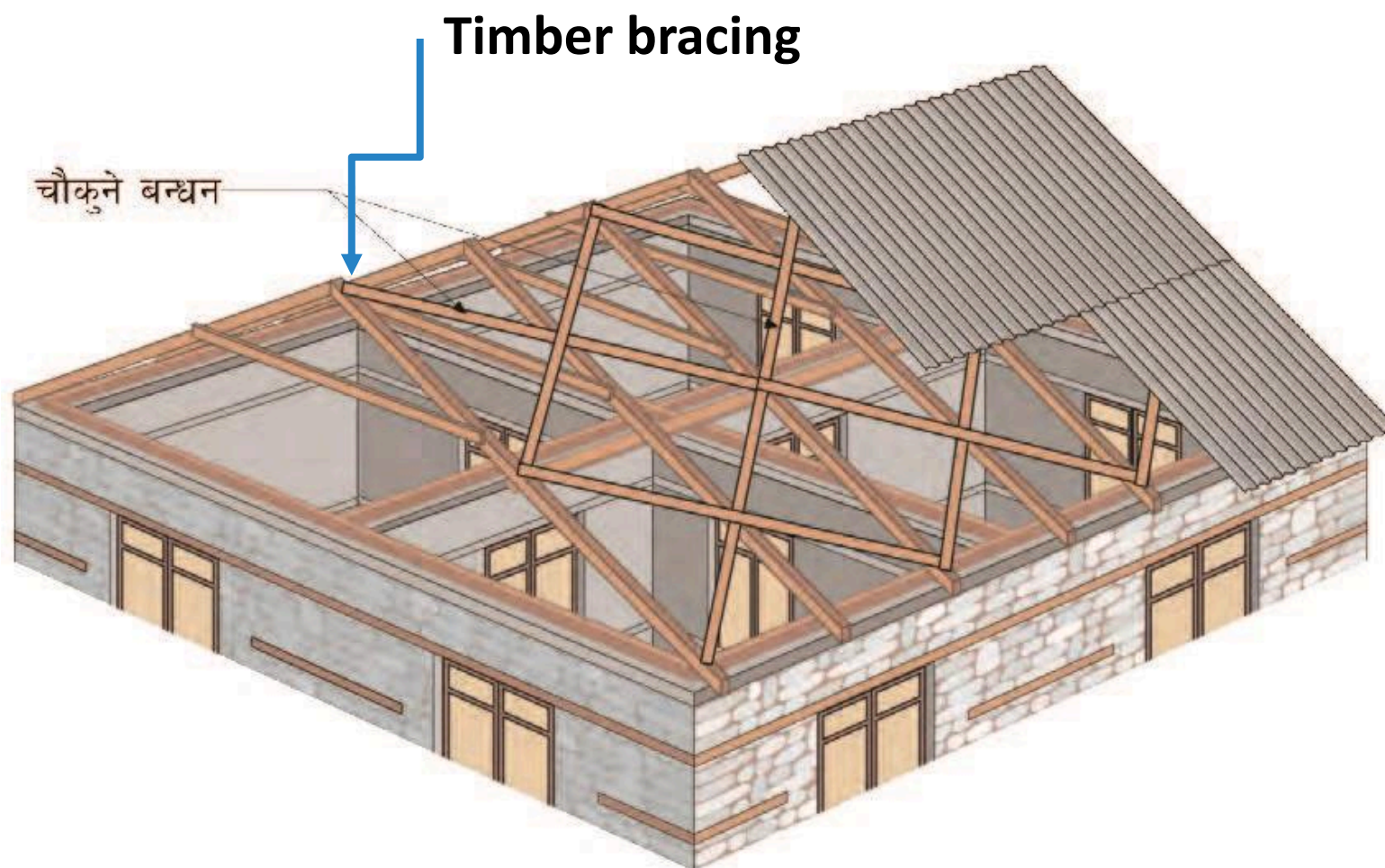


Half cut bamboo overlay



7.0 Stiffening of roofs in their plane

Option 1: Timber Bracing

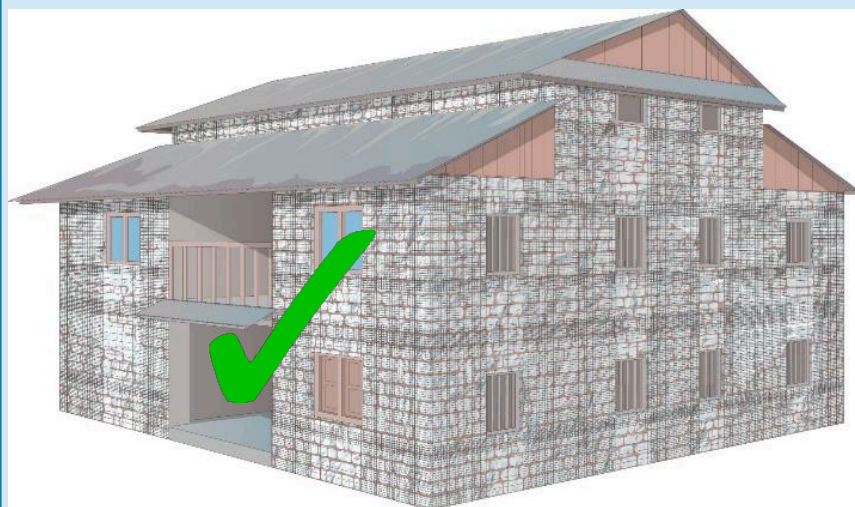


छानामा चौकुने बन्धनको विवरण

8.0 Tying of parapet walls

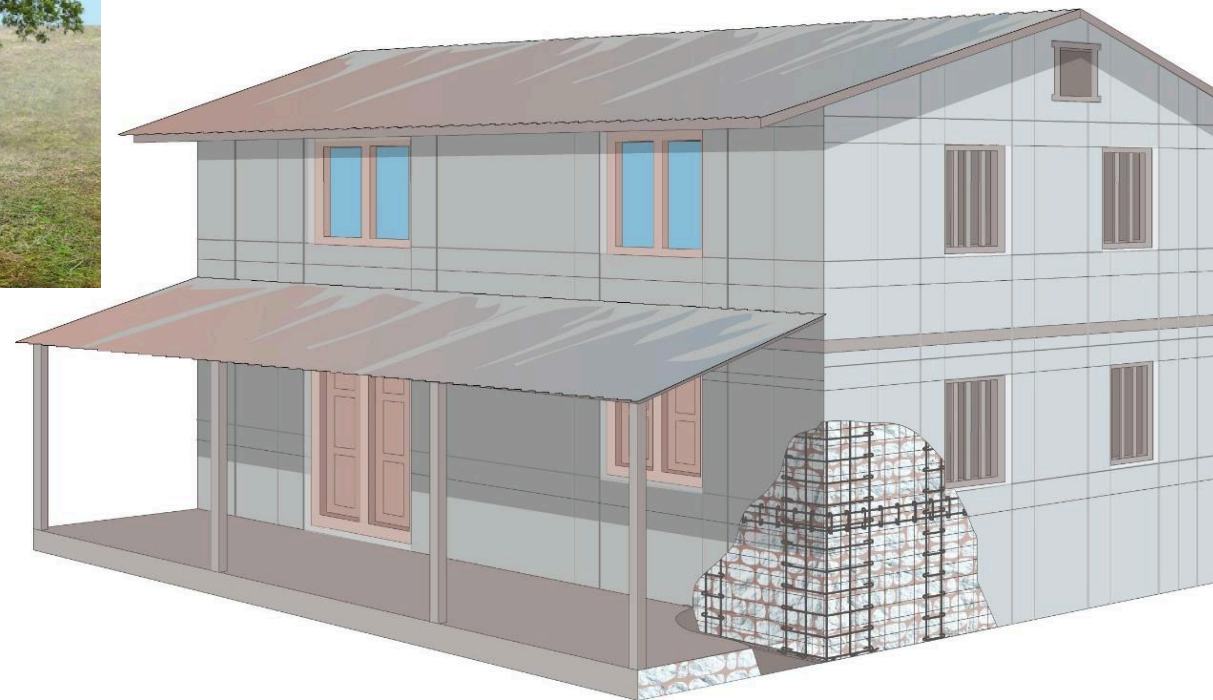


9.0 Tying of gable walls...



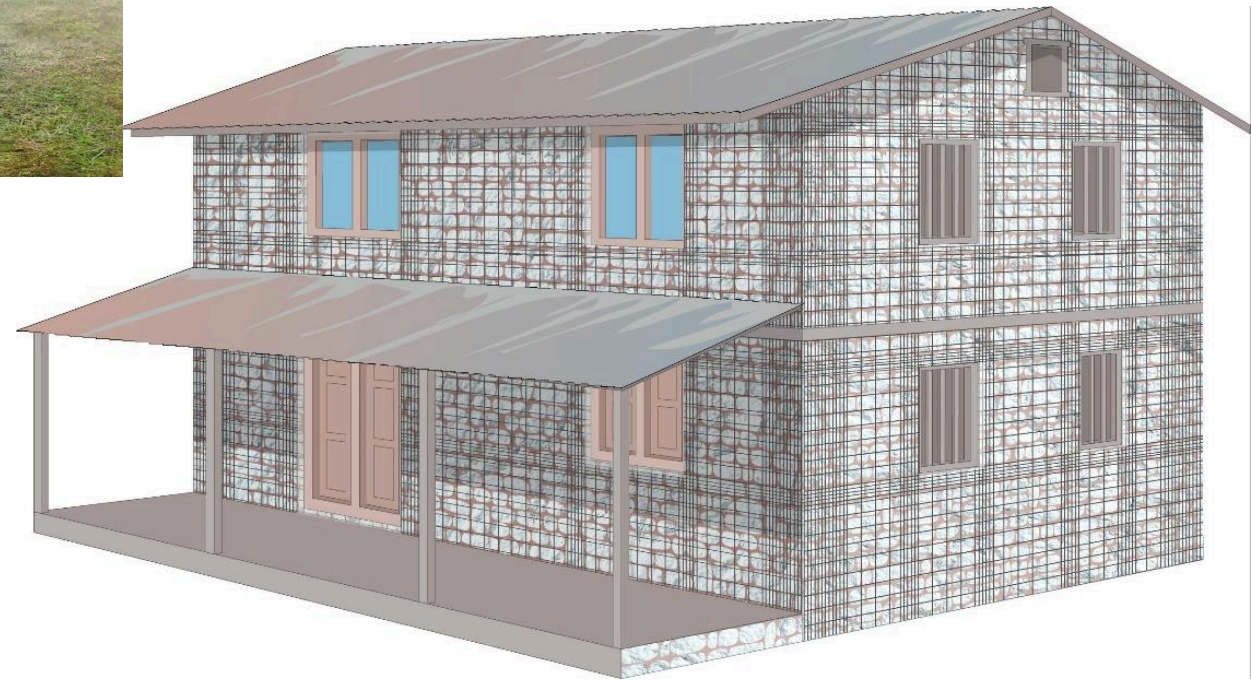
Capacity improvement of structural wall

Contd.



Option 1 :Retrofitting using RC splint-bandage and GI wires in remaining part

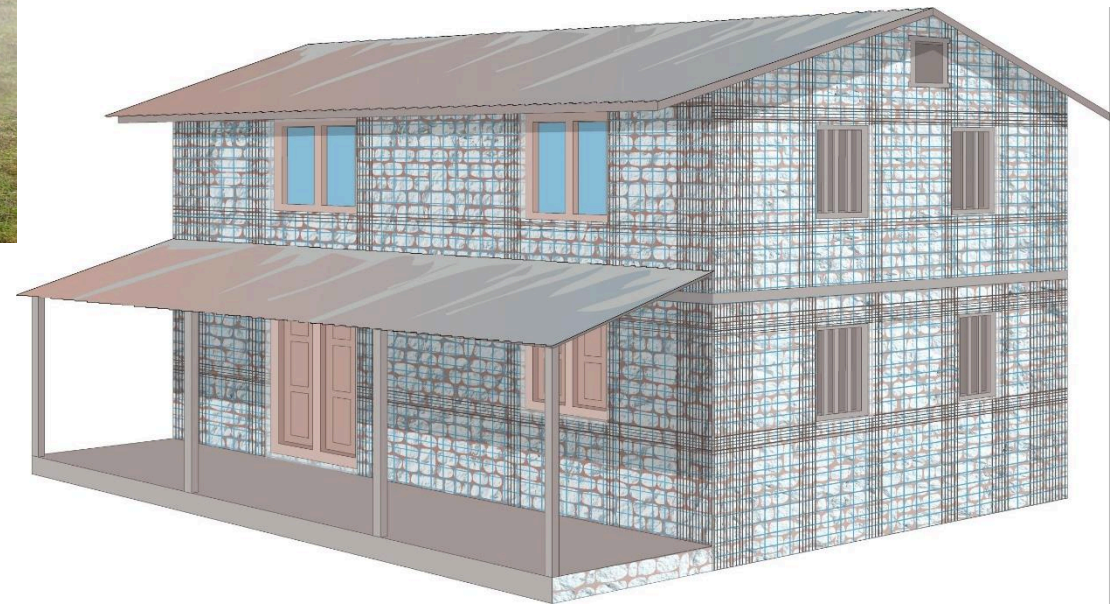
Capacity improvement of structural wall



Option 2: Retrofitting using Welded wire mesh splint-bandage and GI wire mesh Jacketing

Capacity improvement of structural wall

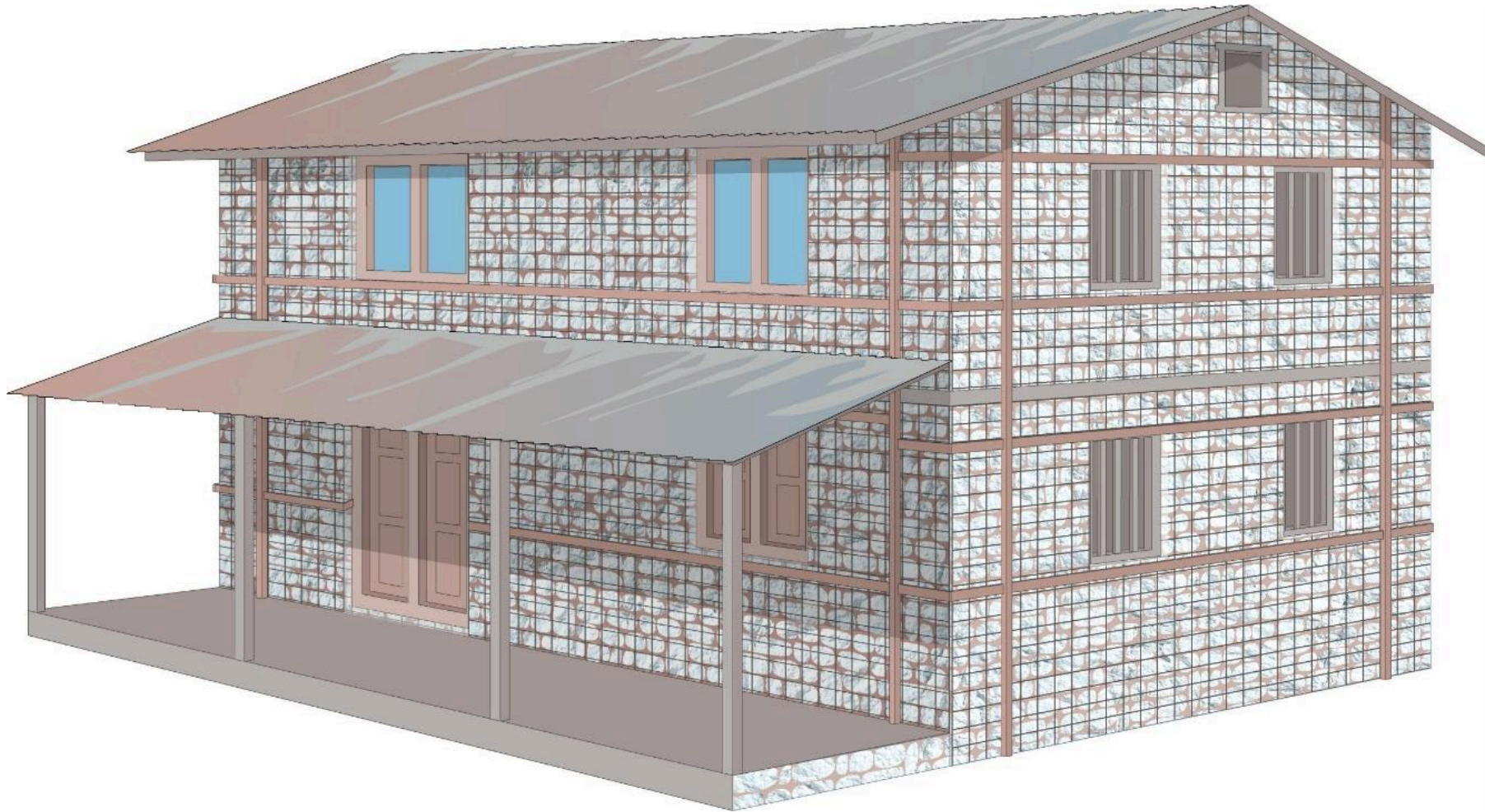
Contd.



Option 3: Retrofitting using Welded wire mesh splint and P.P. wire mesh Jacketing

Capacity improvement of structural wall

Contd.



Option 4 : Retrofitting using wooden section and GI wire mesh Jacketing

PART-C : Ready to use seismic retrofit designs

This section presents summary of retrofit designs which are applicable in following cases :

❖ Retrofitting Stone masonry building

- 1) Retrofitting stone masonry building in mud (RSMM)
- 2) Retrofitting dry stone masonry building (RDSM)

❖ Retrofitting brick masonry building

- 1) Retrofitting brick masonry building in mud with flexible floor (RBMM)
- 2) Retrofitting brick masonry building in cement with flexible floor (RBMC1)
- 3) Retrofitting brick masonry building in cement with rigid floor (RBMC2)

Typical description of building



- Number of storey : 2 plus attic (maximum), except RBMC2 which is three storey
- Storey height : 3.00 m (maximum)
- Total height : 7.0m (2 plus attic) and 9.0 m (three storey)
- Unsupported wall length: 5.40 m (maximum)
- Plinth area : 100.00 sq.m.
- Configuration and load path : is similar as mention in part B : Seismic deficiency and intervention)
- Redundant : Yes
- Material condition : Good or replaced with new material in case of damaged.

Option 1 : Summary of design : For RC splint & bandage



Table 3: Summary of retrofit design (applicable to RSMM, RDSM, RBMM RBMC1 : 2 plus attic storey & RBMC-2* : three storey)

S.N.	Length or Wall	Rebars Reinforcement in seismic belts with overlapping of Ld mm	
	In meter	Concrete Size (mm)	Rebar (No & diameter)
1.	≤ 5.40	300 x 40	2#10

Note : Material grade : M20 and Fe 500 or 415 , ties 4.75 mm diameter bars @ 150 m spacing.

Split : Rebar in RC seismic splint with overlapping of Ld mm,									
SN	No. of storey	Storey	Concrete size	At T-Junction		At Corner Junction		At near opening	
Concrete Grade M 20, Rebar Grade Fe 500				No	Bar ϕ (mm)	No	Bar ϕ (mm)	No	Bar ϕ (mm)
1	One		200x40	3	8	3	8	2	8
2.	One plus attic	attic	200x40	3	8	3	8	2	8
		Ground	200x40	3	8	3	8	2	8
3.	Two	First	200x40	3	8(10)	3	8	2	8
		Ground	200x40	3	8 (10)	3	8	2	8
4.	Two plus attic/second	Attic/Second	200x40	3	10	3	8	2	8
		First	200x40	3	10	3	8	2	8
		Ground	200x40	3	10	3	8	2	8

Option 2 & 3 : Summary of design : For Welded G.I. Mesh splint & bandage



Table 1: Summary of retrofit design (applicable to RSMM, RDSM and RBMM)

S.N. Length or Wall G.I. Mesh Reinforcement in seismic belts with overlapping of 300mm

In meter Gauge No. Wide

1. ≤ 5.40 10 12 300

Gauge, G10 = 3.25mm.

Split : G.I. Mesh Reinforcement in seismic splint with overlapping of 300mm,(applicable to RSMM, RDSM and RBMM)

SN	No. of storey	Storey	G	At T-Junction		At Corner Junction		At near opening	
				No	W	No	W	No	Wide
1	One		12	29 (10)	2X450 +T (450)	19 (10)	450 +T (450)	7 (7)	300 (300)
2.	One plus attic	attic	12	29 (10)	2X450 +T (450)	19 (10)	450 +T (450)	7 (7)	300 (300)
		Ground	12	29 (10)	2X450 +T (450)	19 (10)	450 +T (450)	7 (7)	300 (300)
3.	Two	First	14	54 (17)	2X450 +T (450)	36 (17)	450 +T (450)	9 (9)	200 (200)
		Ground	14	54 (17)	2X450 +T (450)	36 (17)	450 +T (450)	9 (9)	200 (200)
4.	Two plus attic	attic	10	56 (18)	2X450 +T (450)	36 (18)	450 +T (450)	12 (12)	300 (300)
		First	10	56 (18)	2X450 +T (450)	36 (18)	450 +T (450)	12 (12)	300 (300)
		Ground	10	56 (18)	2X450 +T (450)	36 (18)	450 +T (450)	12 (12)	300 (300)

Values in parenthesis () is for splint in side of the room and remaining value is for splint outside the room.

Gauge, G10 = 3.25mm, G12 = 2.64mm

Note : provide PP band or G.I. wires (G12) at 100 mm at centers vertically and horizontally to prevent local failures in remaining portion of the walls.

Option 4 : Summary of design : For wooden splint & bandage



Table 2: Summary of retrofit design (applicable to RSMM and RDSM)

S.N.	Length or Wall	Wooden member in seismic belts with proper overlapping.		
	In meter	size	No.	Wide
1.	≤ 5.40	38mmX75mm	2 (each face)	

Note : Connection using MS plate.

Split : wooden member in seismic splint with proper overlapping .						
SN	No. of storey	Storey	Size	At T-Junction	At Corner Junction	At near opening
				No	No	No
1	One		75x75	4	3	4
2.	One plus attic	attic	75x75	4	3	4
		Ground	75x75	4	3	4
3.	Two	First	75x75	4	3	4
		Ground	75x75	4	3	4
4.	Two plus attic	attic	75x75	4	3	4
		First	75x75	4	3	4
		Ground	75x75	4	3	4

Note : Connection using MS plate. Provide G.I. wire mesh of 12 gauge @100 mm at centres in horizontally and vertically to prevent local failures.

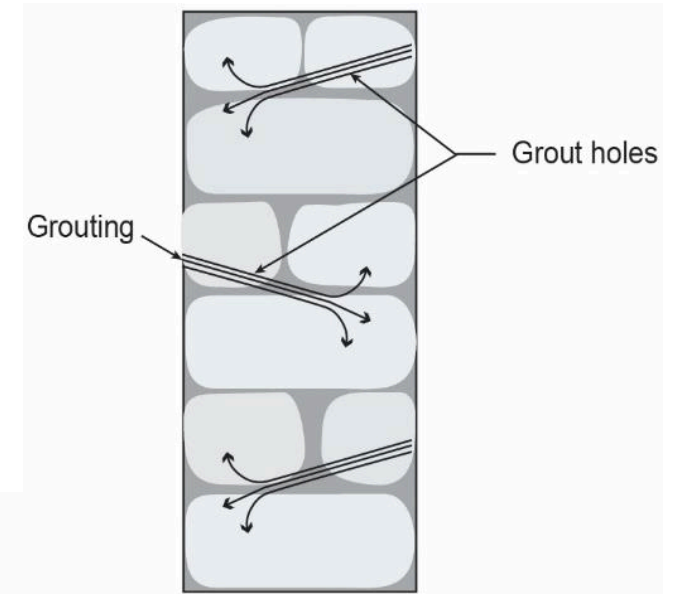
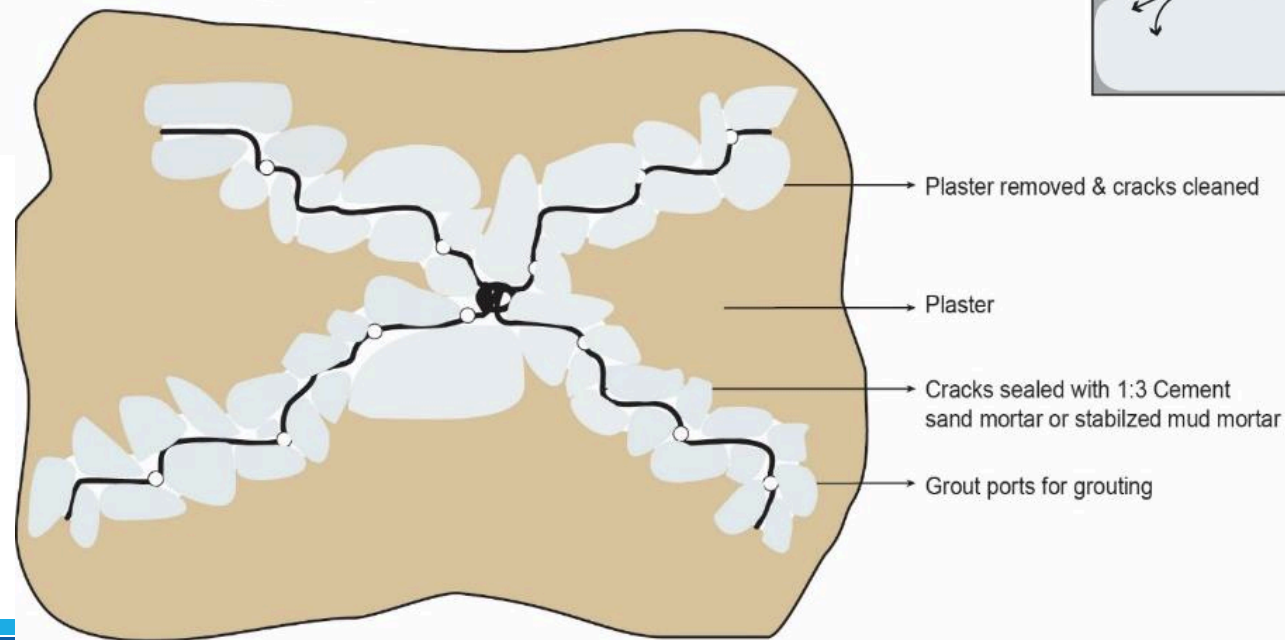
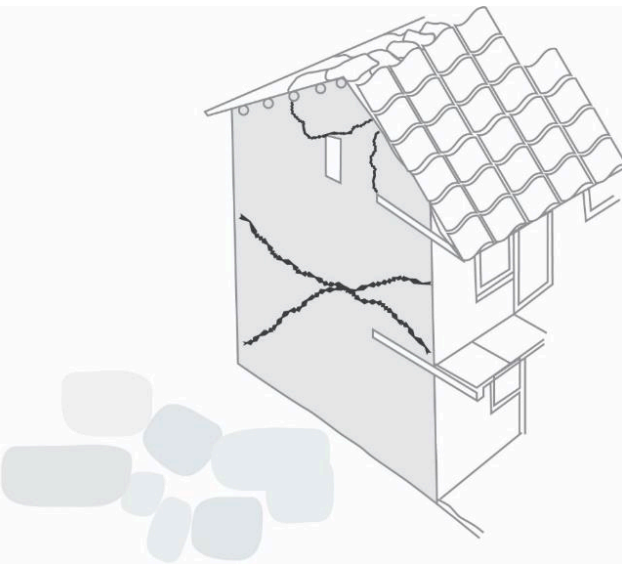
Repair Process

- a)Action # 1:** repair (G1) minor to cracks using grouting
- b)Action # 2:** repair major cracks (G2 to G3) by fixing wire mesh
- c)Action # 3 :** repair major cracks (G2 to G3) by using stitching elements
- d)Action #4:** repair of damaged wall (G2 to G3) by rebuilding

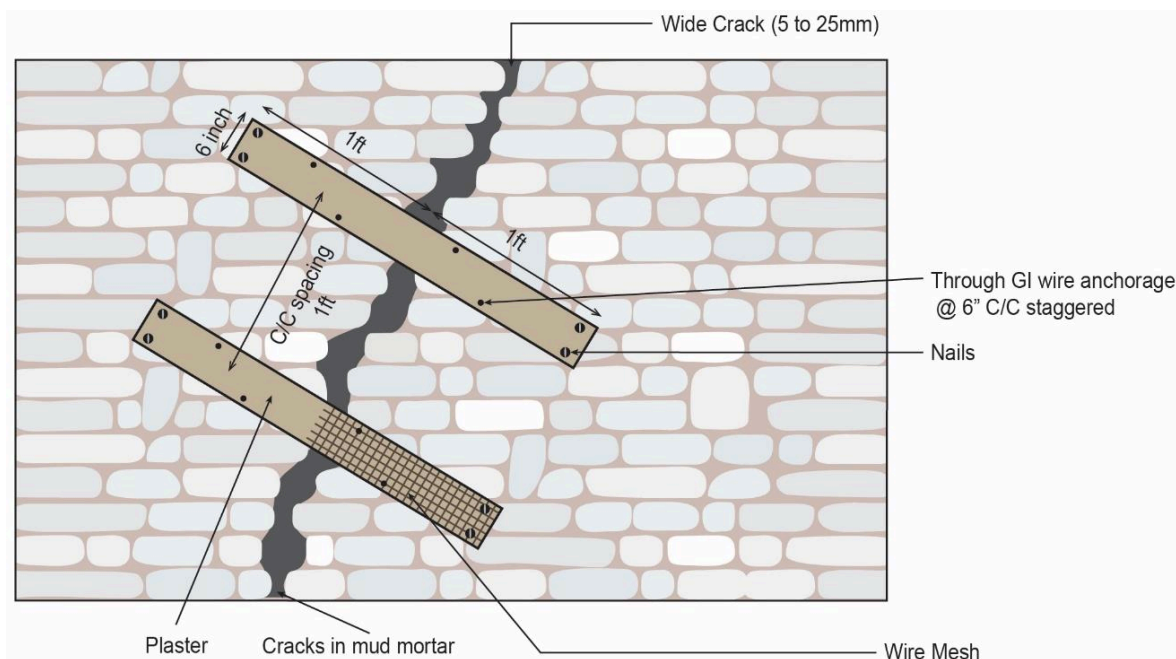
Action # 1: repair (G1) minor to cracks using grouting



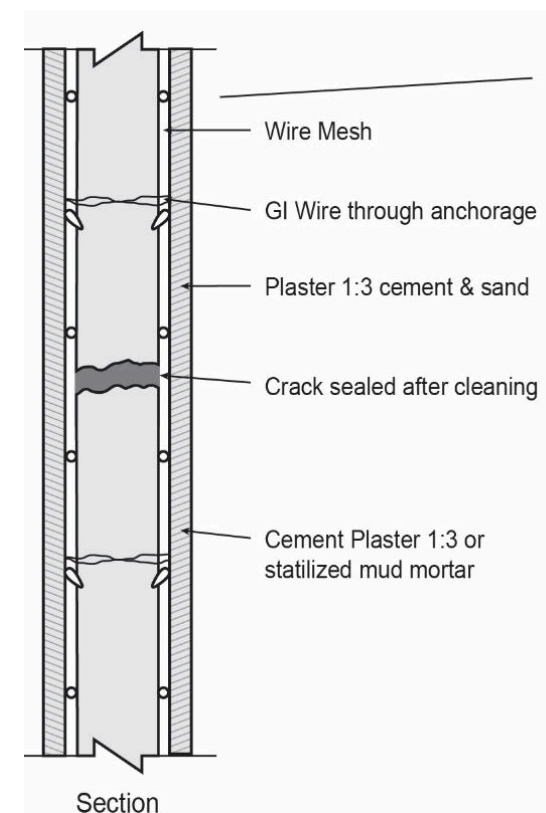
➤ Cracks 0.5 mm to 5mm



Action # 2: repair major cracks (G2 to G3) by fixing



Stone Masonry : Elevation

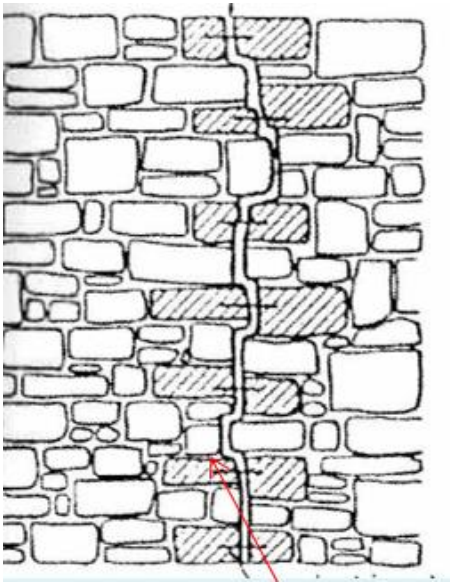


Stone Masonry : Section View

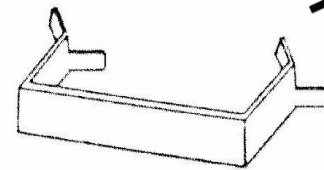
Action # 3 : repair major cracks (G2 to G3) by using stitching elements



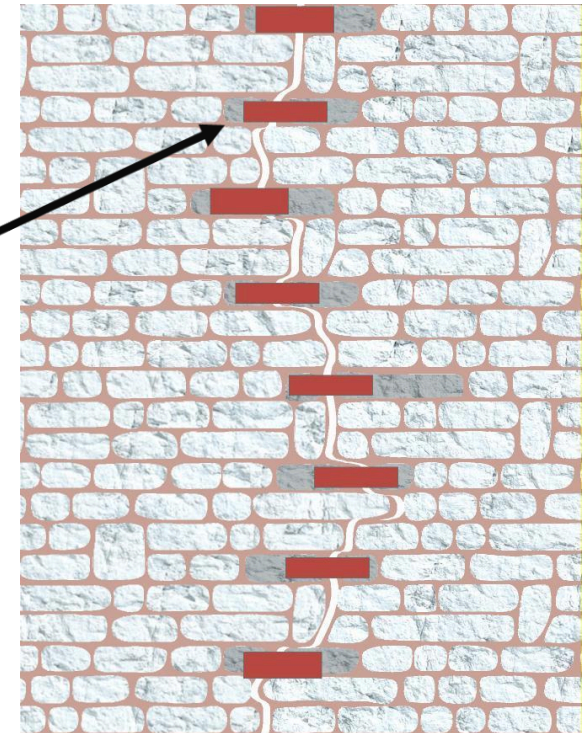
➤ Cracks greater than 5 mm



Removal of stones



Stitching dog (source)



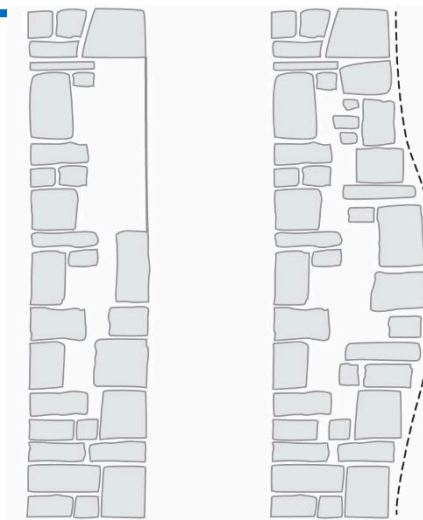
Stone Masonry

Stitching up the crack by inserting stitch on both side of wall

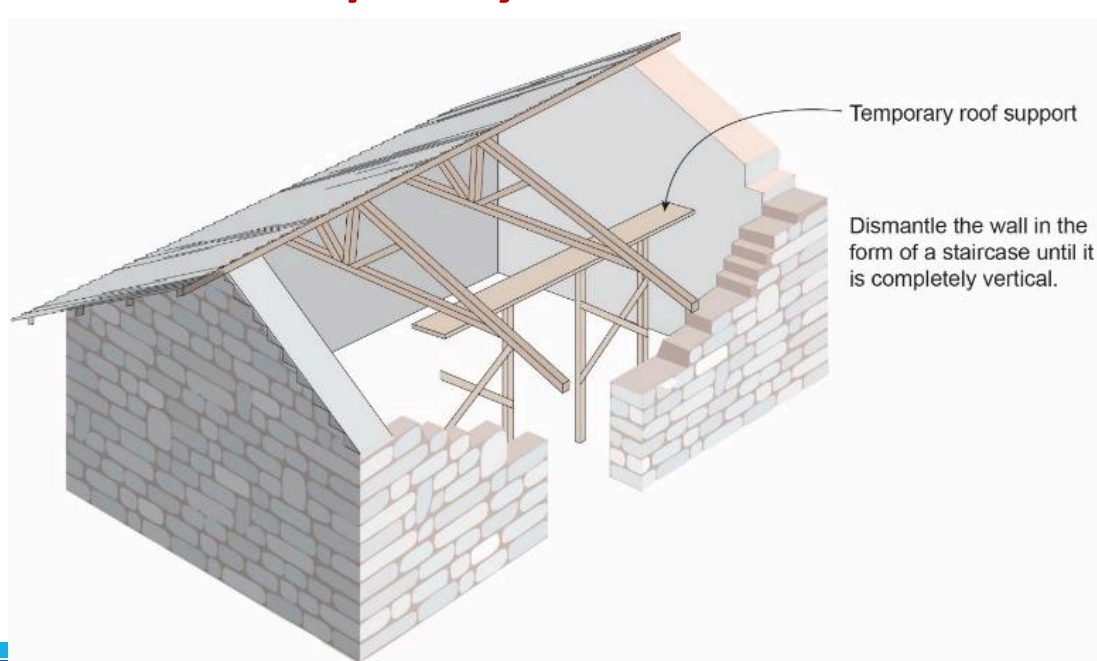
Action #4: repair of damaged wall (G2 to G3) by rebuilding



Stone Masonry

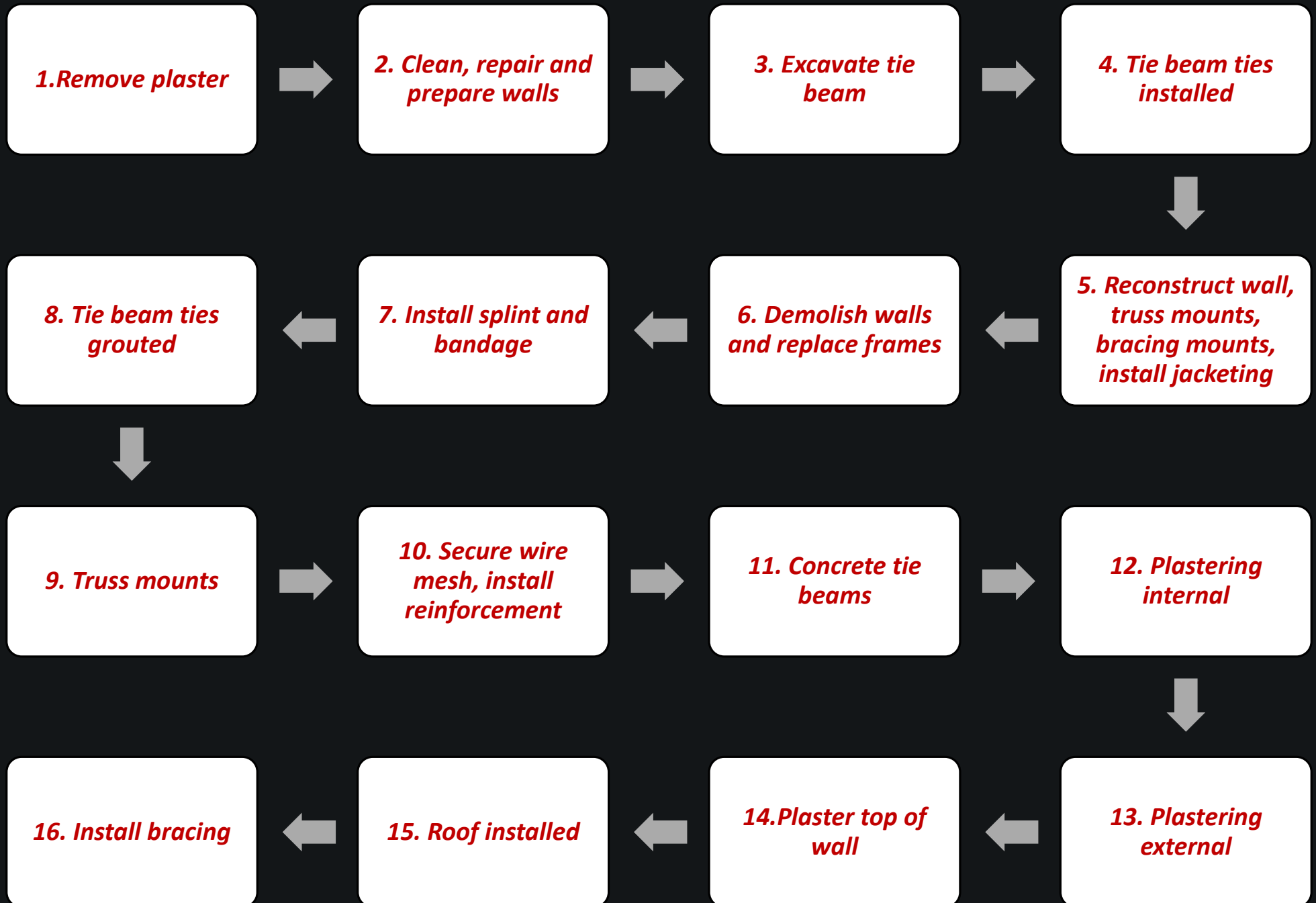


Wall that need completely removed and reconstruction



Retrofitting process

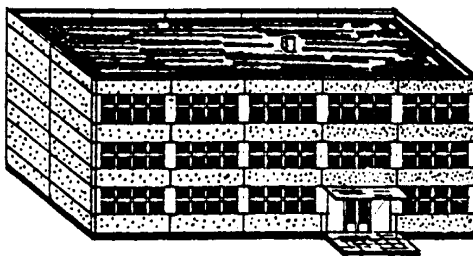
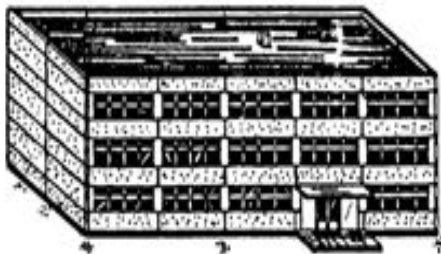
Retrofitting process



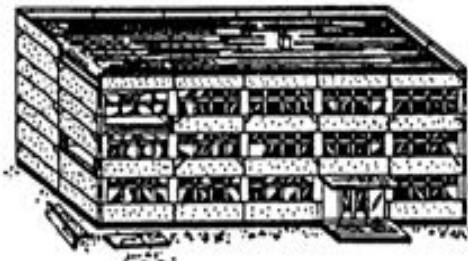
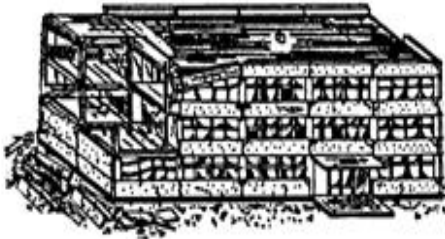
**REPAIR AND
RETROFITTING MANUAL
For
RCC STRUCTURE**

Damage Grades: RC Frame Buildings



Damage Grades			Extent of Damage
DG1	Negligible-slight damage (no structural, slight non-structural)		Hair-line cracks in plaster over frame members or in walls at the base, Fine cracks in partitions and infill
DG2	Moderate damage (slight structural, moderate non-structural damage)		Cracks in columns and beams of frame and in structural walls, Cracks in partition and infill walls, fall of brittle plaster and cladding, falling mortar from joints of wall panel

Damage Grades: RC Frame Buildings

Damage Grades			Extent of Damage
DG3	Substantial to heavy Damage (moderate Structural, heavy non-structural damage)	 A black and white line drawing of a three-story rectangular building. The drawing shows significant damage to the structure, including large cracks in the beams and columns, and some concrete spalling. The building is shown from a slightly elevated perspective.	Cracks in beam and column at the base, spalling of concrete covers, buckling of steel bars, Large cracks in partitions and infill walls, failure of individual infill panels
DG4	Very heavy damage (Heavy structural , very heavy non- structural damage)	 A black and white line drawing of a three-story rectangular building. The drawing shows severe damage to the structure, including large cracks in the beams and columns, and some concrete spalling. The building is shown from a slightly elevated perspective.	Large cracks in structural elements with compression failure of concrete and fracture of rebars, bond failure of beam bars, tilting of columns, collapse of few columns or single upper floor

Damage Grades: RC Frame Buildings



Damage Grades			Extent of Damage
DG5	Destruction (Very heavy structural damage)		Collapse of ground floor or parts of the building

The Provision mentioned in this manual are set as only a advisory measures. For Retrofitting option of RCC Building detail Structural analysis required.

PART-A: Seismic Damage and Intervention	PART-B: Seismic Deficiencies and Intervention
1. Material for repair and retrofitting 2. Basic Repair Techniques for RCC buildings 3. Floor Damage at Ground floor and Mitigation Work [Key problem and repair solution] 4. RCC beam and column damage and Mitigation works [Key problem and repair solution] 5 RCC Floor Slab Damage and Mitigation works [Key problem and repair solution] 6. Other Crack Repair Technique of structural element 5. Infill Wall damage and Mitigation works [Key problem and repair solution]	1. Major seismic Deficiencies of RCC Building 2. Structural Assessment Checklist- 3. Strengthening of Footing of RCC Building 4. Strengthening of Column and beam 5. Strengthening of slab/diaphragm 6. Strengthening of infill walls 7. Annex1:EMS Damage Grade for RCC buildings 8. References

1. Materials for the Repair and Retrofitting



Ordinary material	Advanced material
➤ Cement Slurry ➤ Cement Mortar ➤ Expansive Cement ➤ Quick setting Cement ➤ Gypsum Cement ➤ Steel Reinforcement ➤ GI wires ➤ Rolled Steel sections	➤ Epoxy resin/epoxy Mortar ➤ Polymer Modified Cementitious Products ➤ Fiber Reinforcement Polymers.

2. Basic Repair techniques



1. Repair of spalling or falling or minor cracks in RC members by cement slurry or mortar
2. Routing and Sealing
3. Stitching
4. Applying External Stress
5. Cement slurry Grouting/Epoxy-injection Grouting
6. Surface Overlay/ shotcrete or guniting
7. Removing buckled and yielded Reinforcement and adding new reinforcement
8. Infill wall repair

RCC Beam and Column Damage and Mitigation works



[Key Problem]

Potential damages in foundation are as follows :

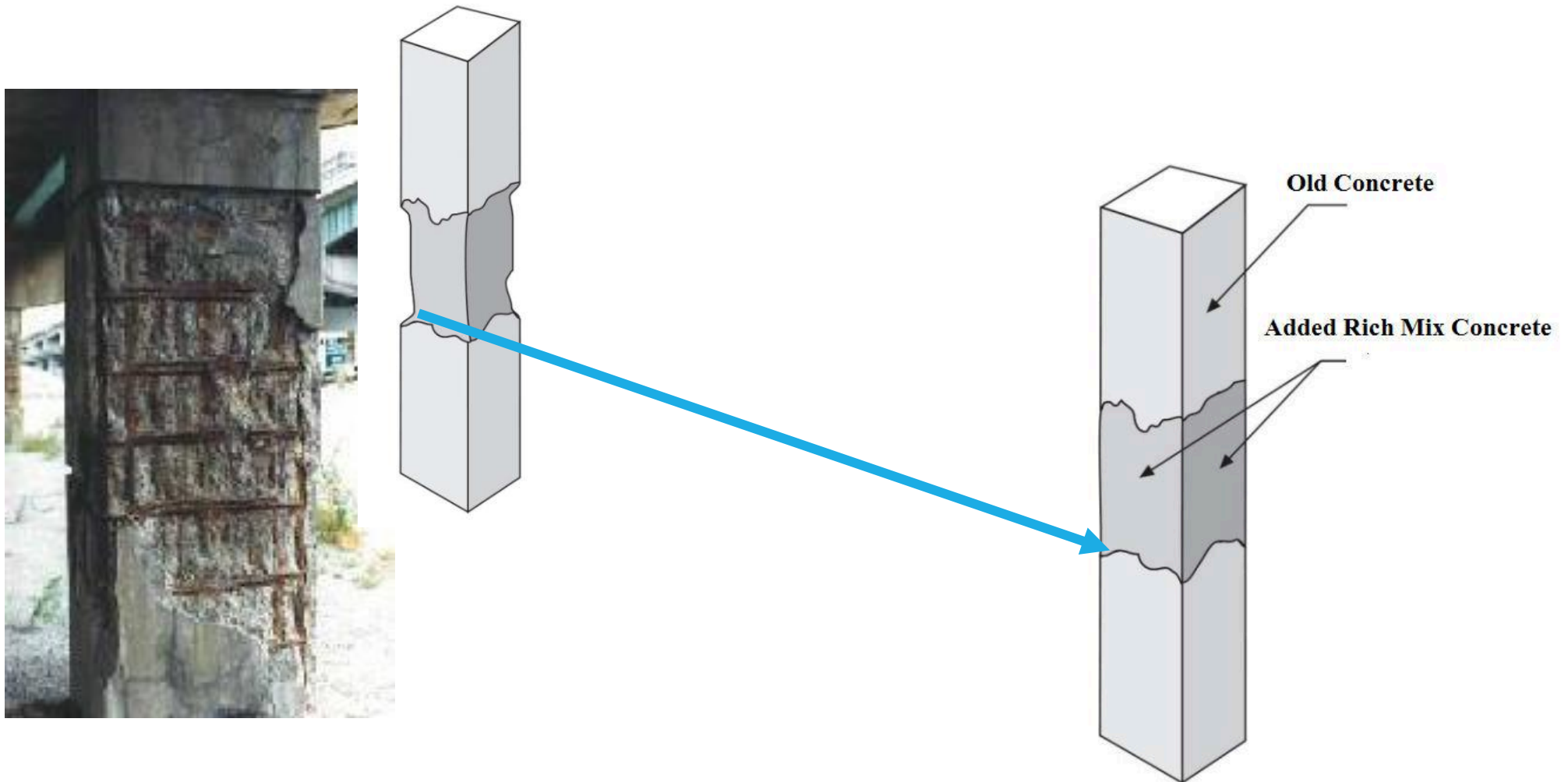
- P.1 spalling or falling of outer cover of concrete
- P.2 minor structural cracks in column and beam
- P.3 Crushing of concrete with or without buckling of bars

[Repair Solution]

Repair solution on corresponding damages in foundation listed above, are :

- S.1 Rich Concrete overlay:
- S.2 Epoxy Grouting in urban and semi-urban areas:
- S.3 Removing buckled and yielded reinforcement and adding new
- Reinforcement:

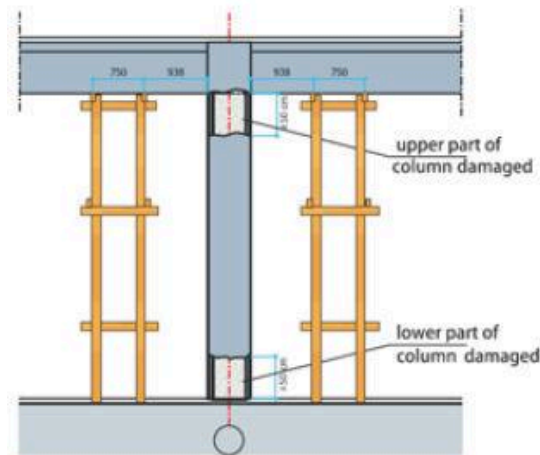
Rich mix concreting Overlay



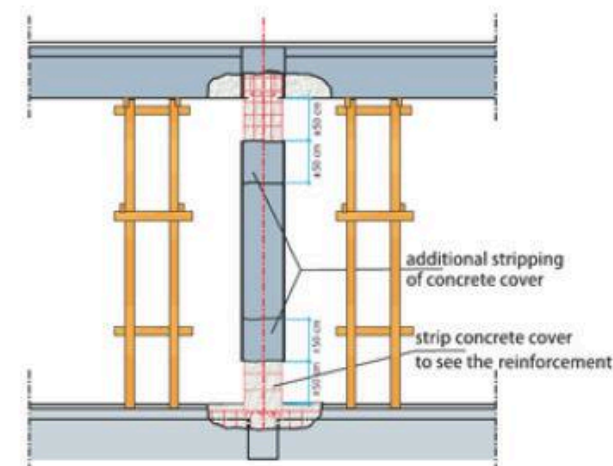
Retrofitting of top and bottom of column that are damaged



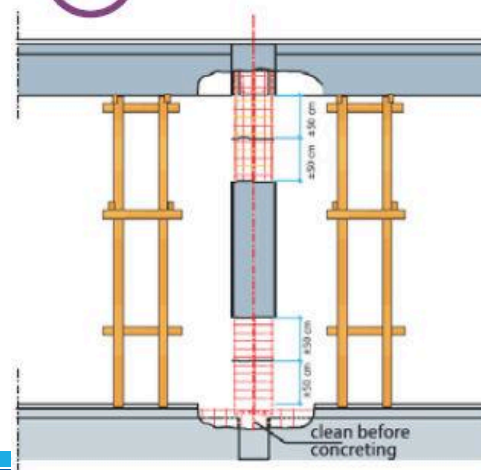
I



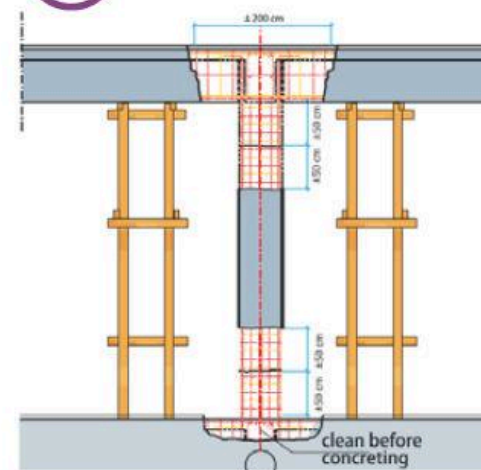
II



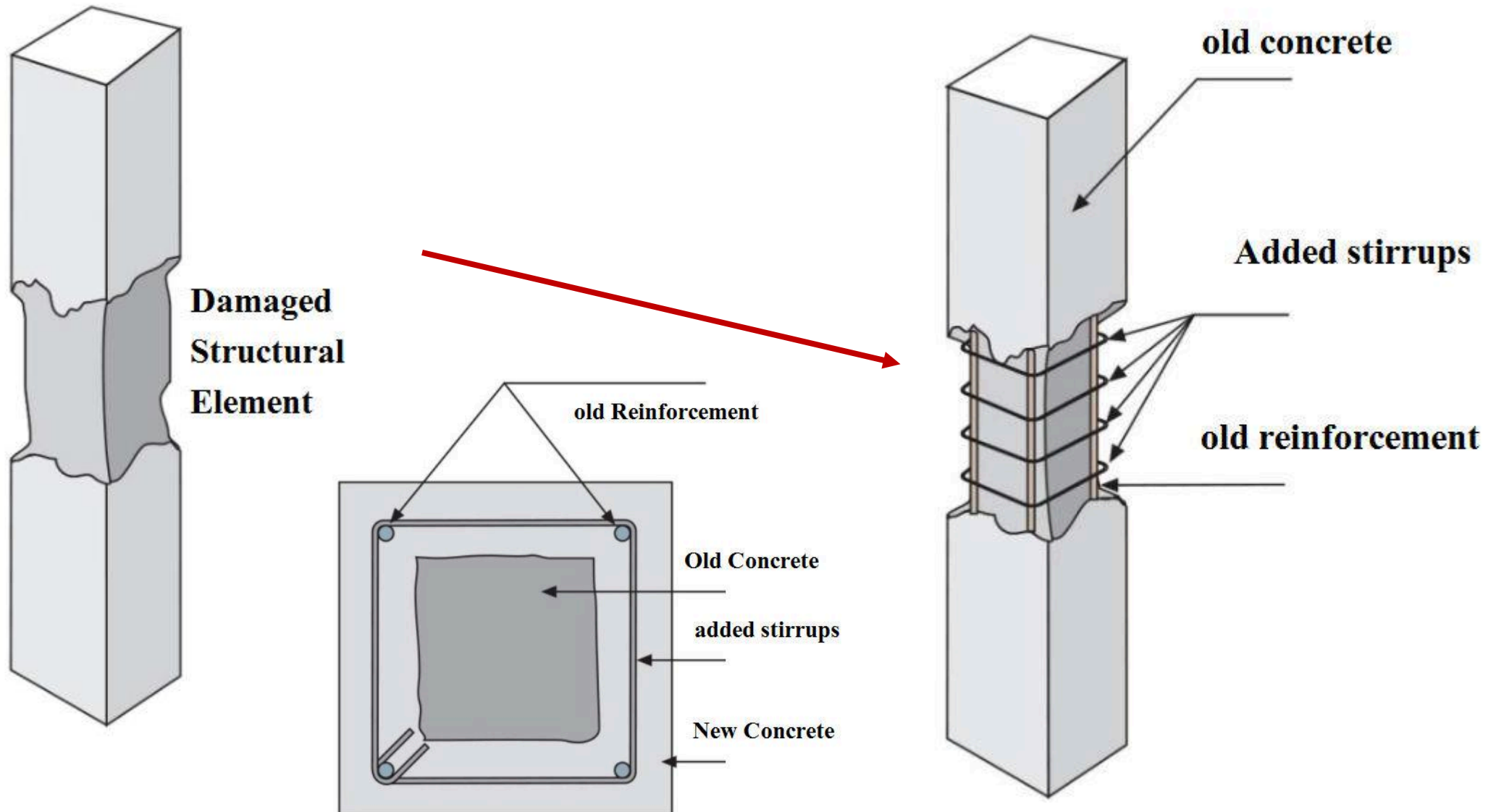
III.A



III.B



retrofitting of column by adding stirrups/reinforcement



PART-B : Seismic deficiencies and intervention

1. Major Seismic Deficiencies of RCC Building



- 1) Inadequate depth and size of the footing
- 2) Inadequate size of Structural Members
- 3) Inadequate amount of reinforcement on structural member
- 4) Missing of bands on masonry walls
- 5) Vulnerable Parapet walls/staircase
- 6) Soft/ weak Storey

- Improvement for Foundation footing and providing
- retaining wall
- Intervention for Beam and column
- Intervention for slab
- Intervention for Masonry infill.

2. Structural Assessment Checklist



Checks	Remarks
Load Path	The frame system provides a complete load path which transfer all inertial forces in the building to the foundation. Is there clear load path?
Goemetry	Is the plan of the building is same in all stories except at basement?
Redundancy	Is There are more than two bays of frame in each direction?
weak/ soft storey	Is there are weak/soft storey?
Vertical discontinuous	Is vertical elements in the lateral force resistant element system are continuous to the foundation except for the basement floor?
Mass	Is there change in mass in adjacent floor of the building?
Torsion	Is the eccentricity of the building within the limit?
Adjacent Building	Are there adjacent building?
Short Column	Is there short column effect?
deterioration of concrete	Is there deterioration of Concrete?

3. Strengthening of Footing of RCC Building

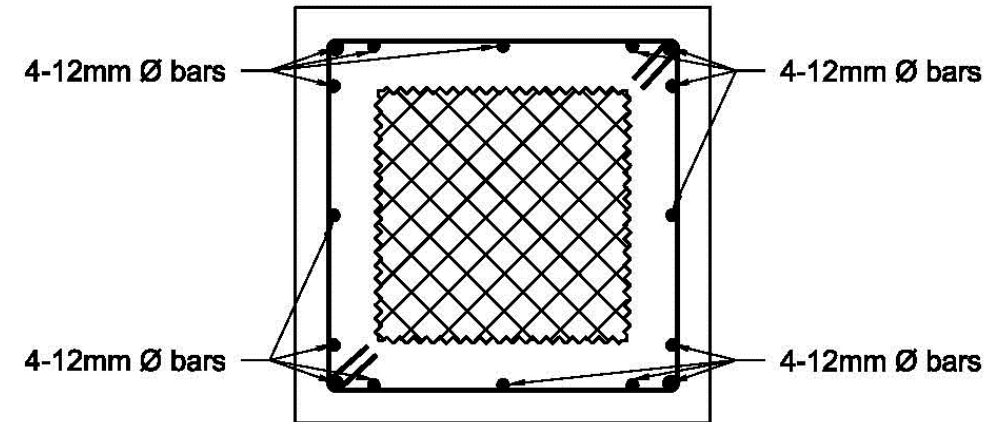


5. Strengthening of Column and beam



1. Provide the required supporting system to the structure.
2. Remove weak concrete if exist.
3. Clean the surface and clean the rust of steel if exist.
4. Apply rust removers and rust preventers.
5. Provide additional steel **as per design** all around the section.
6. Provide required formwork.
7. Provide polymer based bonding coat or rich cement bonding between old and new concrete.
8. Place the concrete of required thickness and grade and workability admixed with plasticizers

Concrete Jacketing



**REINFORCEMENT FOR
COLUMN JACKETING**

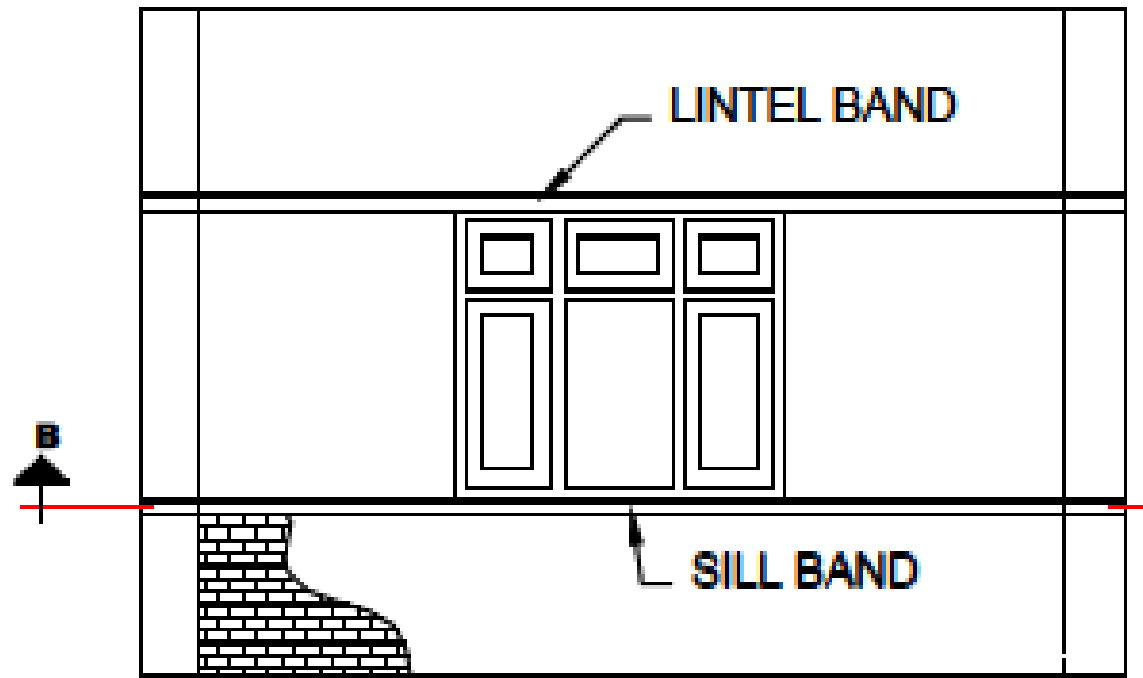
Rolled Steel Jacketing



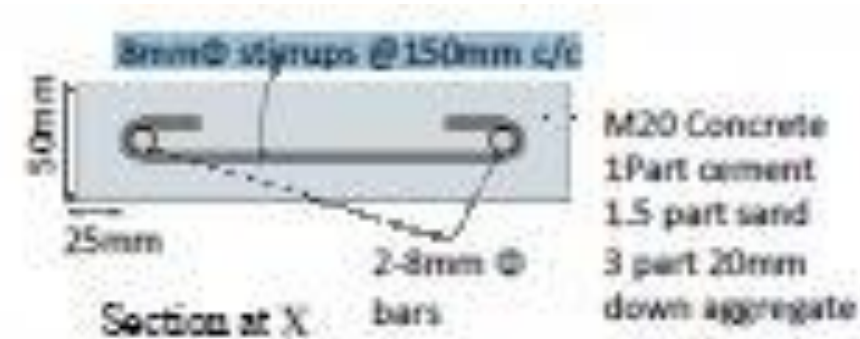
6. Strengthening of Slab/Diaphragm



7. Strengthening of infill walls

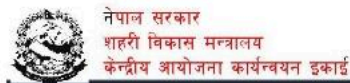


ELEVATION



Adding sill and lintel band externally on masonry wall

अनुसूची - १७ (क) : प्रवलिकरण पूर्व प्राविधिक निरीक्षण तथा प्रमाणीकरण फाराम (प्राविधिक निरीक्षण फारम -१



निरीक्षण फाराम (प्राविधिक निरीक्षण फारम -१)									
घरधनी/लाभग्राहीको जानकारी		निरीक्षण मिति :		गते	महिना	वर्ष			
नाम :		अनुदान सम्झौता नं.		-	-	-	-	-	-
ठेगाना :	जिल्ला: गा.वि.स./न.पा.:	वडा:टोल:		जग्गाको किता नं.:					
फोन / मोबाइल नं.:-		अनुदान सम्झौतामा उल्लेखित बैंक खाता नं.-				बैंकको नाम :-			
खण्ड -१ : घरको जाँचको लागि दिइएको आवेदनमा भएको विवरण									
		आवेदन दर्ता नं.		आवेदन दर्ता मिति.					
		निर्माण सामग्री र प्राविधि							
		छाना र सामग्रीको निर्माण							
प्राविधिक सहायक		☐ छ ☐ छैन	संस्था	☐ नेपाल सरकार ☐ गैरसरकारी संस्था ()					
तालिम प्राप्त ढकमी प्रयोग गरिएको पहिला पुनर्निर्माणमा परेको र पछि प्रवलिकरण गर्न चाहिँने		☐ छ ☐ छैन							
खण्ड -२ : विस्तृत damage and technical suggestion प्राविधिक विवरण									
क)	जग्गा क्षति		छ		छैन				
	क्षतिको स्तर		सामान्य		धेरै				
सुभावहरु									
ख)	सँरचनागत अवयवहरुमा क्षति		छ		छैन				
	क्षतिको स्तर		सामान्य		धेरै				
सुभावहरु (पिलर वाला घर)	पिलर								
	बिम								
	बिम र पिलरको जोनि								
	तल्ला								
	छाना								
गाहो वाला घर	गाहो								
	बिम								
	तल्ला								
	छाना								
	संयोजकता (connectivity)								

अनुसूची १७ (ख)-प्रवलिकरण पश्चात् प्राविधिक निरीक्षण तथा प्रमाणीकरण फाराम (प्राविधिक निरीक्षण-२)



निरीक्षण फाराम (प्राविधिक निरीक्षण-२)									
घरधनी/लाभग्राहीको जानकारी		निरीक्षण मिति :		गते	महिना	वर्ष			
नाम :		अनुदान सम्झौता नं.		-	-	-	-	-	-
ठेगाना :	जिल्ला: गा.वि.स./न.पा.:	वडा:टोल:		जग्गाको किता नं.:					
फोन / मोबाइल नं.:-		अनुदान सम्झौतामा उल्लेखित बैंक खाता नं.-				बैंकको नाम :-			
खण्ड -१ : घरको जाँचको लागि दिइएको आवेदनमा भएको विवरण									
स्वीकृत नक्सा-डिजाइन मध्येको भए		आवेदन दर्ता नं.		आवेदन दर्ता मिति.					
		निर्माण सामग्री र प्राविधि							
		छाना र सामग्रीको निर्माण							
प्राविधिक सहायक		☐ छ ☐ छैन	संस्था	☐ नेपाल सरकार ☐ गैरसरकारी संस्था ()					
तालिम प्राप्त ढकमी प्रयोग गरिएको		☐ छ ☐ छैन							
खण्ड -२ : विस्तृत प्राविधिक विवरण									
विवरण	सुभावहरु	सुभावहरु पालना भएको							
क) जग	१..... २..... ३.....	छ		छैन					
ख) सँरचनागत अवयवहरु									
	पिलर	१..... २..... ३.....							
	बिम	१..... २..... ३.....							
	बिम र पिलरको जोनि	१..... २..... ३.....							
	तल्ला	१..... २..... ३.....							

Thank you